

CCDC32 stabilizes clathrin-coated pits and drives their invagination

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The manuscript presents a **valuable** finding that CCDC32, beyond its reported role in AP2 assembly, follows AP2 to the plasma membrane and regulates clathrin-coated pit assembly and dynamics. The authors further suggest that the alpha-helical region of CCDC32 interacts with AP2 via the alpha appendage domain to mediate this function. While live-cell and ultrastructural imaging data are **solid**, future biochemical studies will be needed to confirm the proposed CCDC32-AP2 interaction.

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Abstract

Clathrin-mediated endocytosis (CME) is essential for maintaining cellular homeostasis. Previous studies have reported more than 50 CME accessory proteins; however, the mechanism driving the invagination of clathrin-coated pits (CCPs) remains elusive. We show by quantitative live cell imaging that siRNA-mediated knockdown of CCDC32, a poorly characterized endocytic accessory protein, leads to the accumulation of unstable flat clathrin assemblies. CCDC32 interacts with the α -appendage domain (AD) of AP2 *in vitro* and with full length AP2 complexes in cells. Deletion of aa78-98 in CCDC32, corresponding to a predicted α -helix, abrogates AP2 binding and CCDC32's early function in CME. Furthermore, clinically observed nonsense mutations in CCDC32, which result in C-terminal truncations that lack aa78-98, are linked to the development of cardio-facio-neuro-developmental syndrome (CFNDS). Overall, our data demonstrate the function of a novel endocytic accessory protein, CCDC32, in regulating CCP stabilization and invagination, critical early stages of CME.

Summary

We show that CCDC32, a poorly studied and functionally ambiguous protein, binds to AP2 and regulates CCP stabilization and invagination. Clinically observed mutations in CCDC32 lose their ability to interact with AP2 likely contributing to the development of cardio-facio-neuro-developmental syndrome.

Introduction

Clathrin-mediated endocytosis (CME) regulates nutrient uptake and maintains the activity of transmembrane transporters and is thus, essential for maintaining cellular homeostasis (Kaksonen and Roux, 2018; Kirchhausen et al., 2014; McMahon and Boucrot, 2011; Mettlen et al., 2018). Malfunctions of CME are strongly associated with neurological diseases, cardiovascular diseases, and cancers (Blue et al., 2018; DeMari et al., 2016; Elkin et al., 2015; Gilles Moulay, 2019; Hamdan et al., 2017; Manti et al., 2019; Sznajder and Swanson, 2019; Wu and Yao, 2009). CME occurs via the assembly of clathrin triskelia into clathrin-coated pits (CCPs) that invaginate to form clathrin-coated vesicles (CCVs) (Kaksonen and Roux, 2018; Mettlen et al., 2018; Smith and Smith, 2022). During CCP maturation, successful invagination of the clathrin coat and its underlying membrane is a key step that determines whether nascent CCPs are productive or abortive (Aguet et al., 2013; Baschieri et al., 2020; Chen and Schmid, 2020; Wang et al., 2020). Although more than 50 endocytic accessory proteins (EAPs) have been reported to be involved in the progression of CME (Bhave et al., 2020; Chen and Schmid, 2020; Kirchhausen et al., 2014; McMahon and Boucrot, 2011; Merrifield and Kaksonen, 2014; Sochacki et al., 2017; Taylor et al., 2011; Traub, 2011), the mechanism of CCP invagination remains elusive, i.e. which accessory proteins are required and how do they regulate CCP invagination?

Coiled-coil domain-containing protein 32 (CCDC32), also known as gene *C15orf57*, is a small, 185 amino acids (aa) protein that has been poorly studied and whose function was unknown. Clinical genome sequencing of three patients with cardio-facio-neuro-developmental syndrome (CFNDS) revealed three homozygous nonsense mutations that only express the first 9, 54 and 80aa of CCDC32, respectively (Abdalla et al., 2022; Harel et al., 2020). How these loss-of-function mutations result in CFNDS remains unknown.

A genome-wide co-essential modules study (Wainberg et al., 2021) suggested a functional correlation between CCDC32 and AP2, whose interactions were also implied by affinity purification-mass spectroscopy (AP-MS) analysis (Cho et al., 2022) and co-immunoprecipitation (co-IP) assays (Wainberg et al., 2021). In addition, depletion of CCDC32 was observed to inhibit transferrin uptake, suggesting a role in CME (Wainberg et al., 2021). More recently, and while this paper was under review, Wan et al. (Wan et al., 2024) reported an essential role for CCDC32 as a co-chaperone with AAGAB (alpha and gamma adaptin binding protein) in the assembly of AP2 complexes. The authors showed that in CRISPR-mediated CCDC32 knockout cells, AP2 complexes were severely depleted, and CME was strongly inhibited. They further showed *in vitro*, that CCDC32 was recruited to the AAGAB:α:σ2 hemicomplex, where it displaced AAGAB, recruited μ2 and β2 subunits to assemble the mature AP2 complex and then was released. They were unable to detect interactions between a C-terminally tagged CCDC32 and the mature AP2 complex (Wan et al., 2024).

Using a combination of biochemistry, quantitative live cell imaging and ultrastructure electron microscopy, we report that a functional N-terminally tagged CCDC32 interacts with the intact AP2 complex and is recruited to CCPs. siRNA knockdown of CCDC32 inhibits CCP invagination and stabilization, without impacting the levels of AP2 expression. Thus, we demonstrate a second, critical role for CCDC32 in regulating early stages of CME by regulating the stabilization and invagination of CCPs.

Results

CCDC32 is recruited to CCPs

We first explored a direct role for CCDC32 in CME by testing if and when CCDC32 is recruited to CCPs. ARPE-HPV cells that stably express mRuby-clathrin light chain a (mRuby-CLCa) and siRNA-resistant, N-terminally GFP tagged full-length CCDC32 (eGFP-CCDC32(FL)) were generated (**Fig. S1**) and imaged using Total Internal Reflection Fluorescence Microscopy (TIRFM). In addition to diffuse staining across the inner plasma membrane (PM) surface, suggestive of direct membrane binding, we also observed colocalization of eGFP-CCDC32(FL) and mRuby-CLCa (**Fig. 1A**) at clathrin coated pits.

Furthermore, cohort-averaged fluorescence intensity traces were obtained using time-lapse imaging and primary (mRuby-CLCa)/subordinate (eGFP-CCDC32) tracking powered by cmeAnalysis (Aguet et al., 2013; Chen et al., 2020; Jaqaman et al., 2008). As a negative control, ARPE-HPV cells that stably express mRuby-CLCa and eGFP showed neither diffuse PM staining nor eGFP recruitment to CCPs (**Fig. S2**), whereas more than half of analyzed CCPs were observed to recruit CCDC32. The recruitment curve of CCDC32 follows the assembly of clathrin on CCPs (**Fig. 1B**). Importantly, compared to CCDC32-positive CCPs, those that fail to recruit eGFP-CCDC32 exhibited significantly shorter, and exponentially decaying lifetimes, previously shown to be characteristic of abortive CCPs (**Fig. 1C**) (Aguet et al., 2013). Together these data establish that the early recruitment, along with clathrin, of CCDC32 to nascent CCPs regulates their maturation and lifetimes.

CCDC32 depletion inhibits transferrin receptor (TfnR) uptake and CCP formation

To gain further insight into the function of CCDC32 in CME, we next examined the effects of siRNA-mediated knockdown of CCDC32 on CME and CCP dynamics. In agreement with the previous observation (Wainberg et al., 2021), siRNA-mediated knockdown of CCDC32 in ARPE-HPV cells, which resulted in an ~60% depletion of CCDC32 (**Fig. 2A,B**), reduced the cellular uptake efficiency of TfnR (**Fig. 2C**).

A recent study reported that CCDC32 functions as an essential chaperone for AP2 complex assembly and that CRISPR-mediated knockout of CCDC32 led to a severe decrease in AP2 (Wan et al., 2024). Under our conditions of CCDC32 knockdown, we did not detect any decrease in protein levels of the AP2 complex (**Fig. S3A,B**), suggesting that the residual CCDC32 was fully capable of fulfilling this catalytic function.

To further define which stages of CME were dependent on CCDC32, we knocked down CCDC32 in ARPE-HPV cells that stably express eGFP-CLCa and employed quantitative live-cell TIRFM to visualize and analyze CCP dynamics (Mettlen and Danuser, 2014). The videos and kymographs of time-lapse imaging showed that CCDC32 depletion resulted in the formation of brighter, static clathrin-coated structures (CCSs) (**Fig. 2D,E** & Videos 1,2) that dominate the images. These have been seen under other perturbation conditions (Aguet et al., 2013; Chen et al., 2020) and reflect the accumulation of CCPs that are either larger, flatter or both. However, we also noted a

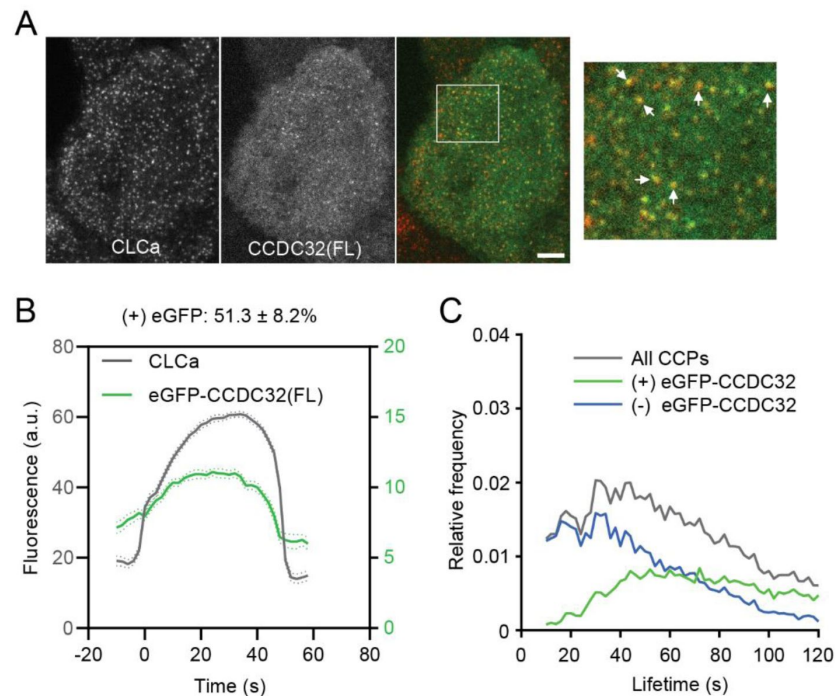


Figure 1

CCDC32 is recruited to clathrin-coated pits.

(A) Representative TIRFM images of ARPE-HPV cells that stably express mRuby-CLCa and eGFP-CCDC32(FL). This dual channel imaging was conducted without siRNA-mediated knockdown. White arrows point to colocalized CLCa and CCDC32 clusters. White ROI is magnified on the right. Scale bar = $5\mu\text{m}$. **(B)** Cohort-averaged fluorescence intensity traces of CCPs (marked with mRuby-CLCa) and CCP-enriched eGFP-CCDC32(FL). $51.3 \pm 8.2\%$ of analyzed CCPs showed eGFP-CCDC32(FL) recruitment. Number of tracks analyzed: 23699. **(C)** Lifetime distributions of all CCPs, CCPs with eGFP-CCDC32 recruitment, and CCPs without eGFP-CCDC32 recruitment.

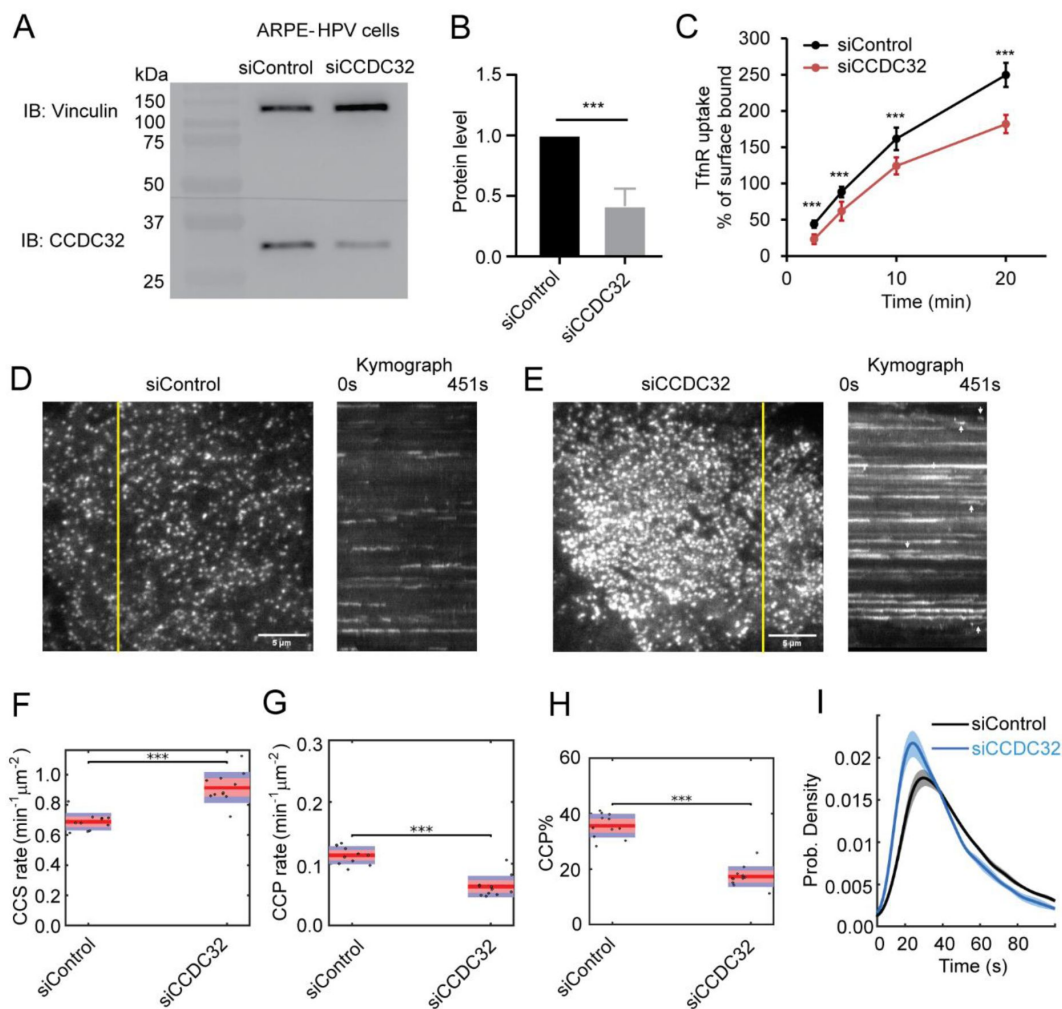


Figure 2

CCDC32 depletion inhibits Transferrin Receptor (TfnR) uptake and CCP maturation.

(A) Immunoblotting (IB) shows efficient CCDC32 knockdown in ARPE-HPV cells by siRNA treatment. (B) Quantified knockdown efficiency (~60%) of CCDC32 (n=8). (C) Measurements of the uptake efficiency of TfnR (n=8). % of surface bound = Internalized/Surface bound*100%. Error bars in (B-C) indicate standard deviations. (D-E) Representative single frame images from TIRFM videos (7.5 min/video, 1 frame/s, see videos 1 and 2) and corresponding kymographs from region indicated by yellow lines of ARPE-HPV eGFP-CLCa cells treated with (D) control siRNA or (E) CCDC32 siRNA. Scale bars = 5 μ m. (F-H) Effect of CCDC32 knockdown on the initiation rates of (F) all CCSs and (G) *bona fide* CCPs, as well as (H) the % of *bona fide* CCPs. Each dot represents a video (n=11). Statistical analysis of the data in (F-H) is the Wilcoxon rank sum test, ***, $p \leq 0.001$. (I) Lifetime distribution of *bona fide* CCPs. Data presented were obtained from a single experiment (n = 11 videos for each condition) that is representative of 3 independent repeats. Number of dynamic tracks analyzed: 125897 for siControl and 105313 for siCCDC32. Shadowed area indicates 95% confidential interval.

subpopulation of dynamic CCPs (arrows, **Fig. 2E**) that were visually obscured by bright static CCSs, yet represent the majority of total CCSs detected. Indeed, despite the deceptive nature of the images, which we have previously reported (Chen et al., 2020), the percentage of static CCSs (lifetime > 150s), as determined by unbiased quantitative analysis, was only 7.9% (**Fig. S4**).

To quantify the dynamic behaviors of CCPs, we performed cmeAnalysis (Aguet et al., 2013; Jaqaman et al., 2008; Loerke et al., 2011) and DASC (disassembly asymmetry score classification) (Wang et al., 2020), which together provide a comprehensive and unbiased characterization of CCP intermediates and CME progression when CCDC32 was depleted. DASC, which measures fluctuations of clathrin-assembly/disassembly to accurately distinguish abortive from productive CCPs (Wang et al., 2020), revealed an increased rate of nascent clathrin assembly, reported as the initiation rate of clathrin-coated structures (CCSs) (**Fig. 2F**), but a reduced rate of *bona fide* CCP initiation (**Fig. 2G**) and a correspondingly lower percentage of stable, *bona fide* CCPs (CCP%) (**Fig. 2H**), which are both brighter and longer-lived than abortive CCPs (Wang et al., 2020). Together these data reveal an early defect in the growth and stabilization of nascent clathrin assemblies leading to an increase in the fraction of abortive CCPs. Importantly, the remaining dynamic population of *bona fide* CCPs exhibited shorter lifetimes upon CCDC32 knockdown (**Fig. 2I**), indicating their more rapid maturation.

Together these results demonstrate that CCDC32 is an important endocytic accessory protein involved in CCP stabilization and maturation. Strikingly, despite the profound effects of CCDC32 depletion on CCP dynamics, the efficiency of TfnR uptake was only marginally affected (**Fig. 2C**). These paradoxical effects are typical for endocytic accessory proteins and indicative of their functional redundancy and/or the induction of compensatory mechanisms, including in this case, the observed increased rate of CCS assembly (**Fig. 2E**) and the more rapid maturation of *bona fide* CCPs (**Fig. 2I**) (Aguet et al., 2013; Bhawe et al., 2020; Wang et al., 2020).

CCDC32 regulates CCP invagination

During CCP maturation, successful invagination of the clathrin coat and its underlying membrane has been identified as a key step that determines the fate of CCPs (Aguet et al., 2013; Wang et al., 2020). To determine whether CCDC32 regulates CCP invagination, we first used Epifluorescence (Epi)-TIRF microscopy to measure CCP invagination in live cells (**Fig. 3A**). In this approach, time-lapse Epi and TIRF fluorescence signals were near-simultaneously acquired for ARPE-HPV eGFP-CLCa cells (**Fig. 3B**) and analyzed using primary (TIRF-channel)/subordinate (Epi-channel) tracking powered by cmeAnalysis (Wang et al., 2020). The resulting Epi and TIRF fluorescence intensity traces of CCPs with similar lifetimes were then aligned and averaged to yield intensity cohorts, which were further log-transformed to give average traces of the invagination depth (Δz) of the CCPs' center-of-mass (**Fig. 3A,C**). Here, we chose to present the ~30s lifetime cohort because they represent the invagination behavior of the most frequent tracks (**Fig. 2H**). CCDC32 knockdown strongly inhibited CCP invagination (**Fig. 3C**), suggesting a key role for CCDC32 in regulating this critical early stage in CME.

To independently validate that CCDC32 regulates CCP invagination, we next examined the clathrin-coated structures (CCSs) at higher resolution using Platinum Replica Electron Microscopy (PREM) (Svitkina, 2017). Strikingly, CCDC32 knockdown resulted in a substantial increase in the number of flat CCSs (**Fig. 4A-C**, pseudo-colored blue in panel B, see also **Fig. S5A-C**) and a corresponding higher occupancy of CCSs on the PM (**Fig. 4E**; **Fig. S5D**), consistent with TIRFM and Epi-TIRF microscopy observations. While the flat CCSs we detected in CCDC32 knockdown cells were significantly larger than in control cells (**Fig. 4D**, mean diameter of 147 nm vs. 127 nm, respectively), they are much smaller than typical long-lived flat clathrin lattices (≥ 300 nm) (Grove et al., 2014). Indeed, the surface area of the flat CCSs that accumulate in CCDC32 KD cells (mean $\sim 1.69 \times 10^4$ nm²) remains significantly less than the surface area of an average 100 nm diameter CCV ($\sim 3.14 \times 10^4$ nm²). Thus, we refer to these structures as 'flat clathrin assemblies' because they are neither curved 'pits' nor large 'lattices'. Rather, the flat clathrin assemblies

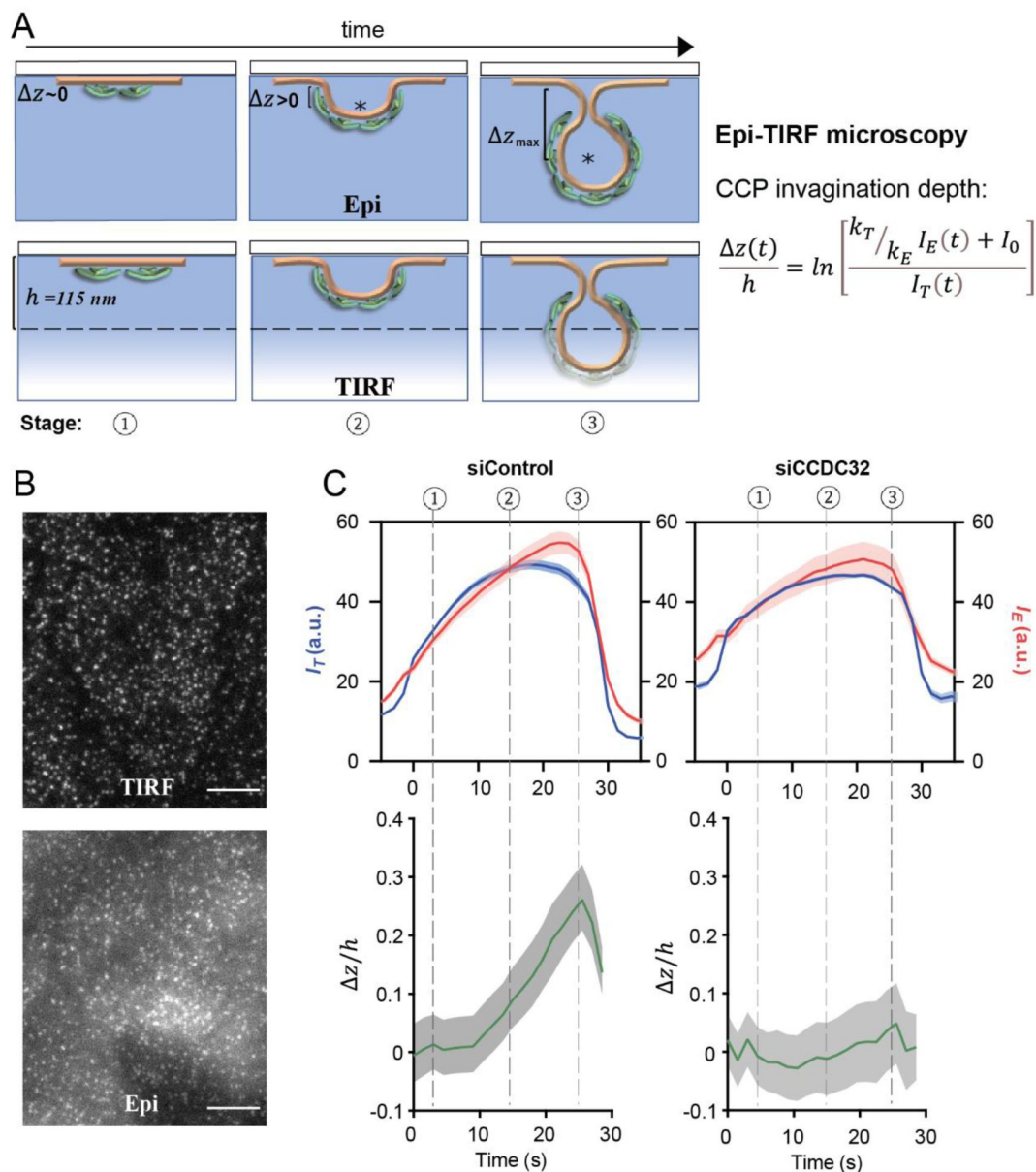


Figure 3

CCDC32 depletion inhibits CCP invagination.

(A) Scheme of Epi-TIRF microscopy for measuring the invagination of CCPs using primary/subordinate tracking. $\Delta z(t)$ denotes the invagination depth of CCPs over time. I_E and I_T denotes the cohort-averaged fluorescence intensity from Epi and TIRF channels, respectively. k_E and k_T are the initial growth rate for the Epi and TIRF channel signals, respectively. I_0 is an additive correction factor. '*' indicates the mass center of clathrin coat. $h=115\text{nm}$ is the evanescent depth of TIRF field, see more detail in ref. (Saffarian and Kirchhausen, 2008; Wang et al., 2020). **(B)** Representative Epi and TIRF microscopy images. Scale bars = $10\mu\text{m}$. **(C)** Epi-TIRF microscopy analysis shows that knockdown CCDC32 strongly inhibited CCP invagination. Top: Cohort-averaged CCP fluorescence intensity traces from Epi and TIRF channels; bottom: calculated $\Delta z(t)/h$ curves. Data presented were obtained from $n = 12$ videos for each condition. Number of CCP tracks analyzed to obtain the $\Delta z/h$ curves: 5998 for siControl and 4418 for siCCDC32. Shadowed area indicates 95% confidence interval.

represent early, likely defective, intermediates in CCP formation. Importantly, while significantly decreased in both number and size, dome-shaped (green) and spherical (orange) CCSs were still detected, likely corresponding to intermediates within the larger subpopulation of dynamic, *bona fide* CCPs able to maintain TfnR uptake (Fig. 4A-D; Fig. 55A-C). The PREM results provide high resolution structural data supporting a critical role for CCDC32 in regulating CCP invagination.

CCDC32 interacts with the AP2 α -appendage domain

Next, we explored the mechanism of CCDC32 recruitment to CCPs. The adaptor AP2, which is essential for CCP and CCV formation, is a heterotetramer consisting of α , β 2, μ 2, and σ 2 subunits (Fig. 5A). A previous study had reported interactions between overexpressed σ 2-mCherry and CCDC32-GFP (Wainberg et al., 2021), but had not demonstrated interactions with the native AP2 complex. A more recent study reported that C-terminally tagged CCDC32 interacts with the α : σ 2 hemicomplex, but does not interact with the assembled AP2 heterotetramer (Wan et al., 2024). To explore this apparent discrepancy, we conducted co-immunoprecipitation (co-IP) experiments from cell lysates of ARPE-HPV cells that stably express a fully functional eGFP-tagged α -subunit of AP2 (Mino et al., 2020) (AP2- α -eGFP, Fig. 5B). Mass Spectrometry analysis revealed, as expected, the co-IP of all the subunits of the AP2 complex, as well as well-known AP2 binding proteins (i.e. EPS15, NECAP2, AAK1 and AAGAB) (Mettlen et al., 2018). Notably, CCDC32 was also efficiently co-immunoprecipitated with intact AP2 (Fig. 5C; Table S1). However, this experiment does not definitively rule out the possibility that CCDC32 only interacts with a small population of immature α : σ 2 hemicomplexes that might be present in the lysate. Therefore, we performed co-IP experiments in cell lysates from ARPE-HPV cells stably expressing eGFP-CCDC32(FL) (Fig. 5D). Western blotting showed that the α subunit (Fig. 5E-F), together with each of AP2 complex subunits (Fig. S3C) were efficiently co-precipitated with eGFP-CCDC32(FL), but not by GFP (Fig. 5D-F). Thus, we conclude that CCDC32 interacts with the mature AP2 complex. The discrepancy with previous findings may be due to the location of the eGFP tag (see discussion).

Most endocytic accessory proteins interact with AP2 via the appendage domains of the α and β 2 subunits (Praefcke et al., 2004; Schmid et al., 2006) (Fig. 5A, α -AD and β -AD). Therefore, we expressed and purified GST-tagged AP2 α -AD and β -AD (Fig. 5G) and used GST pull-down assays to test which might interact with CCDC32. GST-AP2- α -AD, but not GST-AP2- β -AD, was capable of pulling down CCDC32 from ARPE-HPV eGFP-CCDC32(FL) cell lysate (Fig. 5H,I). Thus, CCDC32 can interact with AP2 via the α -AD.

Identification of a CCDC32 binding site for AP2 interactions

We next identified the region on CCDC32 responsible for AP2 binding. Structural predictions of CCDC32 made by AlphaFold 3.0 show it to be a mainly unstructured protein with several isolated α -helices (Fig. S6A,B), which likely led to its misnomer as a coiled-coil domain containing protein. Indeed, even when modelled as a dimer or trimer, we found no evidence of coiled-coil interactions between the α -helices (data not shown). Moreover, endogenous CCDC32 did not co-immuno-precipitate with eGFP-CCDC32 from ARPE-HPV eGFP-CCDC32(FL) cell lysate (Fig. S6C), indicating that the protein is a monomer *in vivo*. Nonetheless, we focused first on a strongly predicted α -helix encoded by aa78-98 and located in the middle of CCDC32 and engineered a CCDC32 construct lacking this α -helix (Δ 78-98, Fig. 6A; Fig. S6A-B). The corresponding siRNA-resistant CCDC32(Δ 78-98) mutant was stably expressed in ARPE-HPV cells at comparable expression levels to eGFP-CCDC32(FL) (Fig. S1A). As previously showed (Fig. S3C) all four subunits of AP2 efficiently co-IP with full length CCDC32 from cell lysates (Fig. 6B,C). Interestingly, the co-IPs of both α and σ 2 subunits with CCDC32(Δ 78-98) were greatly reduced (Fig. 6B,C); whereas the ability of CCDC32(Δ 78-98) to interact with β 2: μ 2 was unaffected. These data are partially consistent with the results of Wan et al. (Wan et al., 2024), who reported interactions with the α : σ 2 hemicomplex and with μ 2, but not with β 2. Given the efficiency and

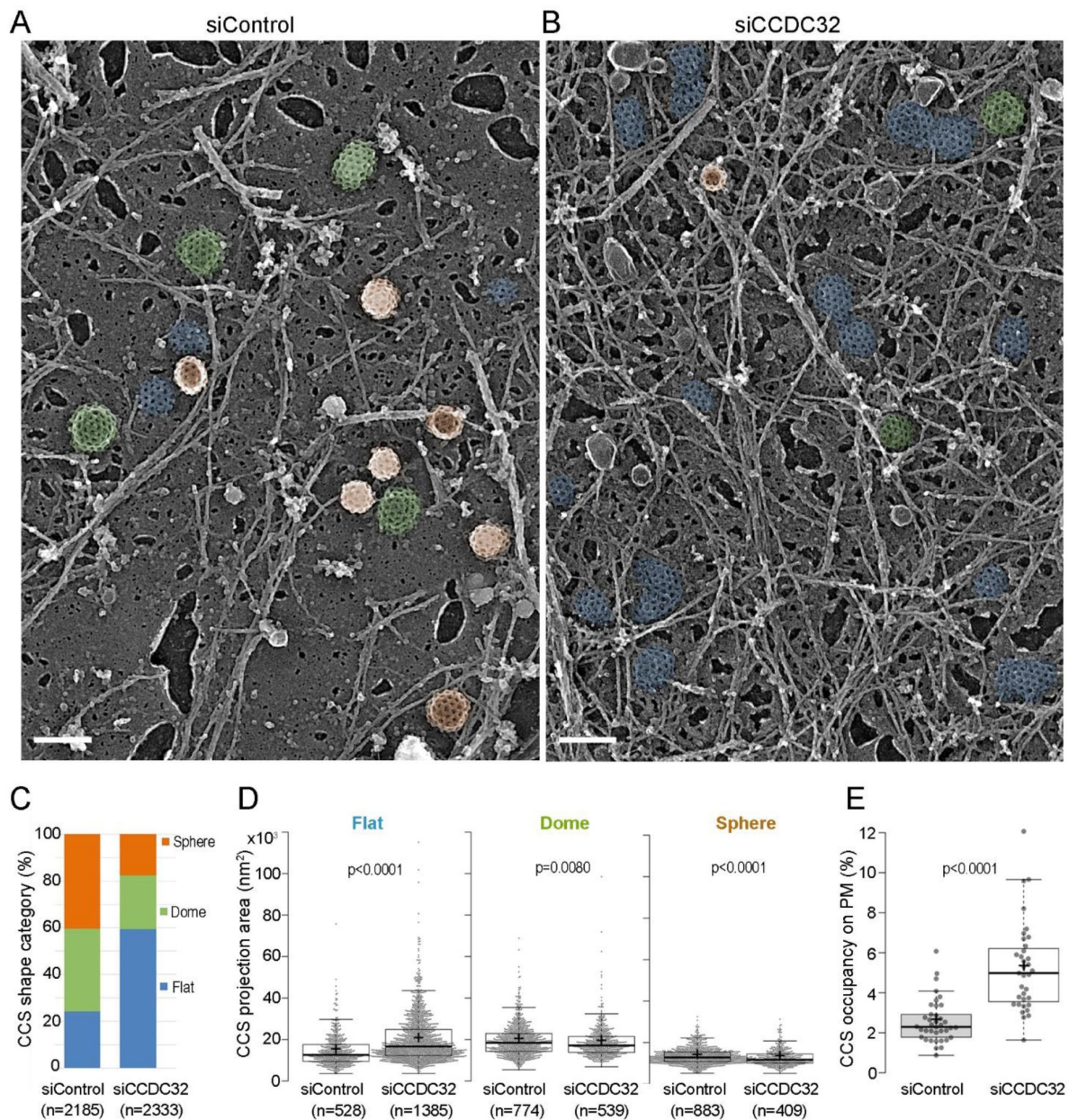


Figure 4

Flat clathrin lattices accumulate in cells depleted of CCDC32.

(A-B) Representative PREM images of ARPE19 cells treated with (A) control siRNA or (B) CCDC32 siRNA showing flat (blue), dome-shaped (green) and spherical (orange) CCSs. Scale bars = 200nm. **(C-E)** Quantification of the clathrin-coated structures (CCSs) by (C) shape category, (D) the CCS projection area, and (E) the CCS occupancy on the plasma membrane (PM). Each dot in (E) represents an individual fragment of the PM; the number of cell membrane fragments analyzed is 38 for siControl cells and 35 for siCCDC32 cells from two independent experiments, n = number of CCS. Statistical tests were performed using Mann-Whitney test (D) or unpaired t-test (E). For the Box and whisker plots in (D) and (E), the box extends from the 25th to 75th percentiles, the line in the middle of the box is plotted at the median, and the “+” indicates the mean.

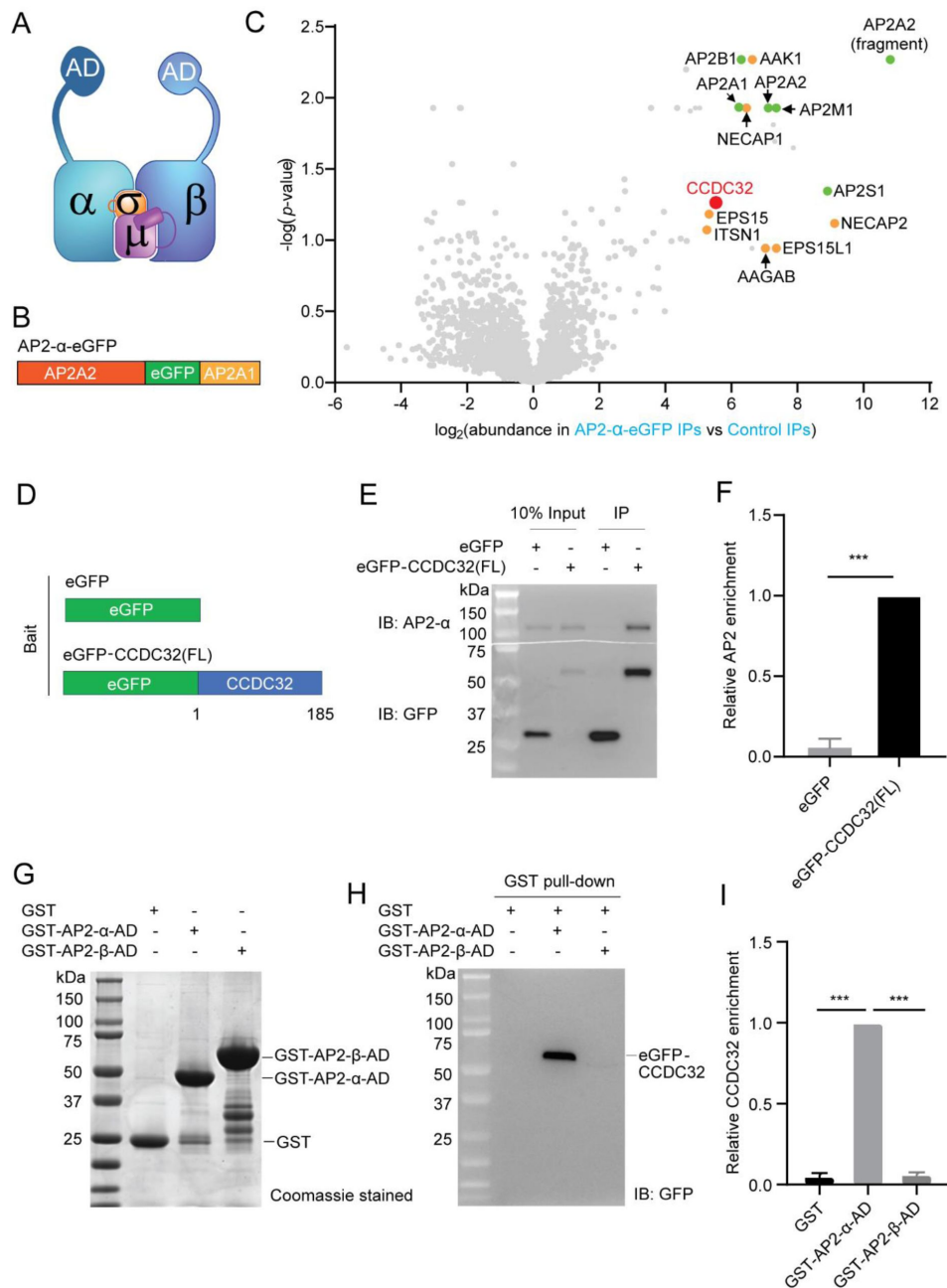


Figure 5

CCDC32 interacts with AP2 through the α appendage domain.

(A) AP2 structure. AD: appendage domain. **(B-C)** Immunoprecipitation (IP) of eGFP from naive ARPE-HPV cells (control IPs) or ARPE-HPV cells that stably express AP2-α-eGFP (AP2-α-eGFP IPs) using anti-GFP beads. **(B)** The domain structure of AP2-α-eGFP. **(C)** Volcano plot after mass spectrometry analysis of the IP samples for protein ID and abundance detection. Green: AP2 subunits; orange: known AP2 interactors. Source data for the volcano plot is available as Supplementary Table 2. **(D-F)** IP of eGFP from ARPE-HPV cells that stably express eGFP or eGFP-CCDC32(FL) using anti-GFP magnetic beads. **(D)** The domain structures of eGFP and eGFP-CCDC32(FL). **(E)** Representative immunoblotting result of n=3 IP samples. **(F)** Relative AP2 enrichments quantified from immunoblotting results. **(G-I)** GST pull-down assays. **(G)** Quantification of immunoblots of the relative enrichment of CCDC32. **(H)** Coomassie blue stained SDS-page gel of purified GST, GST-AP2-α-AD, and GST-AP2-β-AD. **(I)** Representative western blot result of n=3 GST pull-down assay from ARPE-HPV eGFP-CCDC32(FL) cell lysate using purified GST, GST-AP2-α-AD, or GST-AP2-β-AD. The amount of bait GST-proteins is shown in **(D)**, and the pulled down eGFP-CCDC32 was detected by immunoblotting (IB) of GFP. Error bars in **(F)** and **(I)** are SD (n=3). Two-tailed student's t-test: ***, $p \leq 0.001$.

selectivity of our co-IP, our data also suggests that the mature AP2 complex exists in a dynamic equilibrium between $\alpha:\sigma 2$ and $\beta 2:\mu 2$ hemicomplexes. These findings demonstrate that aa78-98 are essential for CCDC32 interactions with both mature AP2 complexes and the $\alpha:\sigma 2$ hemicomplex.

Disrupting AP2 interactions severely impaired the recruitment of CCDC32($\Delta 78-98$) to CCPs (**Fig. 6D,E**), indicating that CCDC32 is likely recruited to CCPs through its interactions with AP2. Notably, unlike CCDC32(FL), CCDC32($\Delta 78-98$) was unable to rescue either the uptake of the CME cargo, TfnR (**Fig. 6F**) or the stabilization of CCPs (**Fig. 6G**). Together, these results establish that CCDC32, a previously uncharacterized endocytic accessory protein, plays a critical role in regulating CCP stabilization and invagination.

Disease-causing nonsense mutation in CCDC32 loses AP2 interaction capacity and inhibits CME

Loss-of-function nonsense mutations in CCDC32 have been reported to result in CFNDS (Abdalla et al., 2022; Harel et al., 2020); however, the disease-causing mechanism remains unknown. The identified frameshift mutations result in premature termination and truncation of the protein at residue 10, 55 or 81 (Harel et al., 2020). Based on our findings, we hypothesize that these C-terminally truncated mutants, all of which lack the α -helix encoded by residues 78-98, will be defective in AP2 binding and unable to function in CME. To test this hypothesis, we generated an siRNA-resistant eGFP-CCDC32(1-54) construct (**Fig. 7A**) based on the clinical report (Harel et al., 2020), and stably expressed this truncated CCDC32 mutant in ARPE-HPV cells (**Fig. S1A**).

As predicted, AP2 did not co-IP with CCDC32(1-54) from the cell lysate (**Fig. 7B,C**), confirming that the disease-causing mutation in CCDC32 abolishes its interactions with AP2. Correspondingly, we could not detect CCDC32(1-54) recruitment to CCPs in dual channel TIRFM imaging (**Fig. 7D,E**), consistent with CCP recruitment of CCDC32 being dependent on CCDC32-AP2 interactions. Interestingly, we also noticed that the level of diffuse plasma membrane binding detected by TIRFM decreased to a greater extent than that seen for CCDC32($\Delta 78-98$), now corresponding to background, eGFP only levels (see **Fig. S7** for direct comparison). These data suggest that the C-terminus of CCDC32 is required for binding to the inner surface of the plasma membrane. Finally, we observed that expression of eGFP-CCDC32(1-54), after siRNA-mediated knockdown of endogenous CCDC32, was unable to rescue TfnR uptake efficiency (**Fig. 7F**) or CCP stabilization (**Fig. 7G**). These findings show that this loss-of-function nonsense mutation in CCDC32 abolishes its interactions with AP2 and inhibits CME likely contributing to the development of CFNDS.

Discussion

CCDC32 was identified through genetic studies, originally in a yeast 2-hybrid screen for proteins interacting with annexin A2 (Li et al., 2011) and subsequently via whole exome sequencing to identify mutations associated with cranio-facio-neurodevelopmental syndrome (CFNDS) (Abdalla et al., 2022; Harel et al., 2020), neither of which provided insight into its cellular function. Subsequent bioinformatic analysis of co-essential modules linked CCDC32 with the AP2 adaptor complex, provided evidence for its interaction with AP2, reported colocalization of CCDC32 with AP2 at CCPs, and demonstrated a role in CME (Wainberg et al., 2021). A more recent report has suggested that CCDC32 functions as a chaperone, essential for the assembly of mature AP2 heterotetrametric complexes (Wan et al., 2024), but that CCDC32 neither binds to the mature AP2 complex nor colocalizes with CCPs.

Here we show that CCDC32 binds to intact AP2 complexes and that this interaction is required for its recruitment to CCPs. Our data demonstrate that CCDC32 plays a critical role at early stages of CME, dependent on its recruitment to CCPs. Depletion of CCDC32 results in a pronounced defect in CCP invagination, a decrease in the rate of formation and percentage of stabilized nascent *bona*

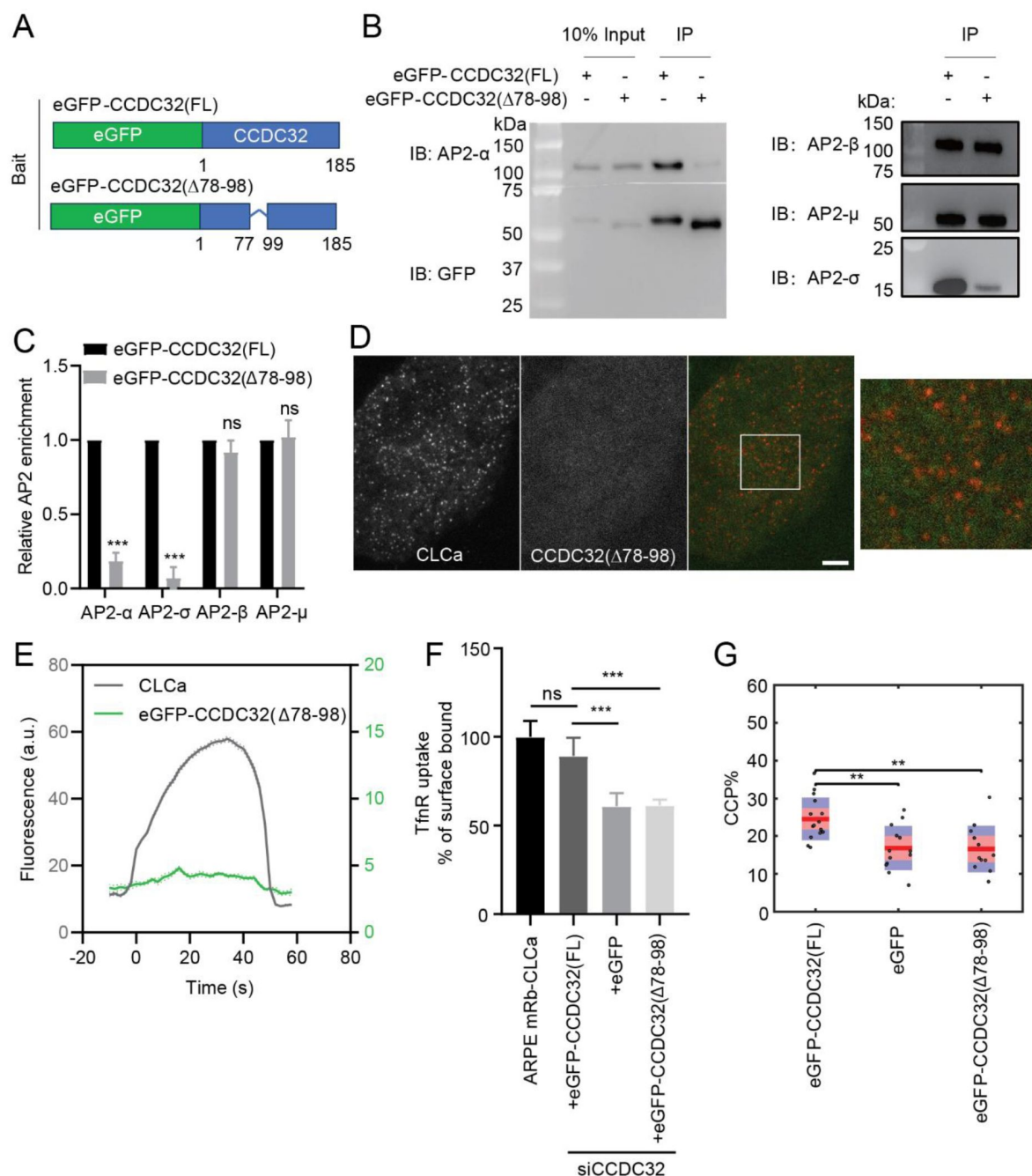


Figure 6

A central α -helix in CCDC32 mediates CCDC32-AP2 interactions.

(A-C) IP of eGFP from ARPE-HPV cells that stably express eGFP-CCDC32(FL) or eGFP-CCDC32(Δ78-98) using anti-GFP beads. (A) The domain structures of eGFP-CCDC32(FL) and eGFP-CCDC32(Δ78-98). (B) Representative Immunoblotting result of n=3 IP samples. (C) Relative AP2 enrichments that were quantified from (B). Error bars indicate standard deviations. **(D)** Representative TIRFM images of ARPE-HPV cells that stably express mRuby-CLCa and eGFP-CCDC32(Δ78-98). White ROI is magnified on the right. Scale bar = 5μm. **(E)** Cohort-averaged fluorescence intensity traces of CCPs and CCP-enriched eGFP-CCDC32(Δ78-98). Number of tracks analyzed: 37892. **(F-G)** (F) TfnR uptake efficiency and (G) CCP% of ARPE mRb-CLCa cells that stably express eGFP-CCDC32(WT), eGFP, or eGFP-CCDC32(Δ78-98), and with siRNA-mediated knockdown of endogenous CCDC32. Each dot in (G) represents a movie. Statistical analysis of the data in (F) is two-tailed student's t-test: ns, not significant; ***, $p \leq 0.001$. Statistical analysis of the data in (G) is the Wilcoxon rank sum test, **, $p \leq 0.01$.

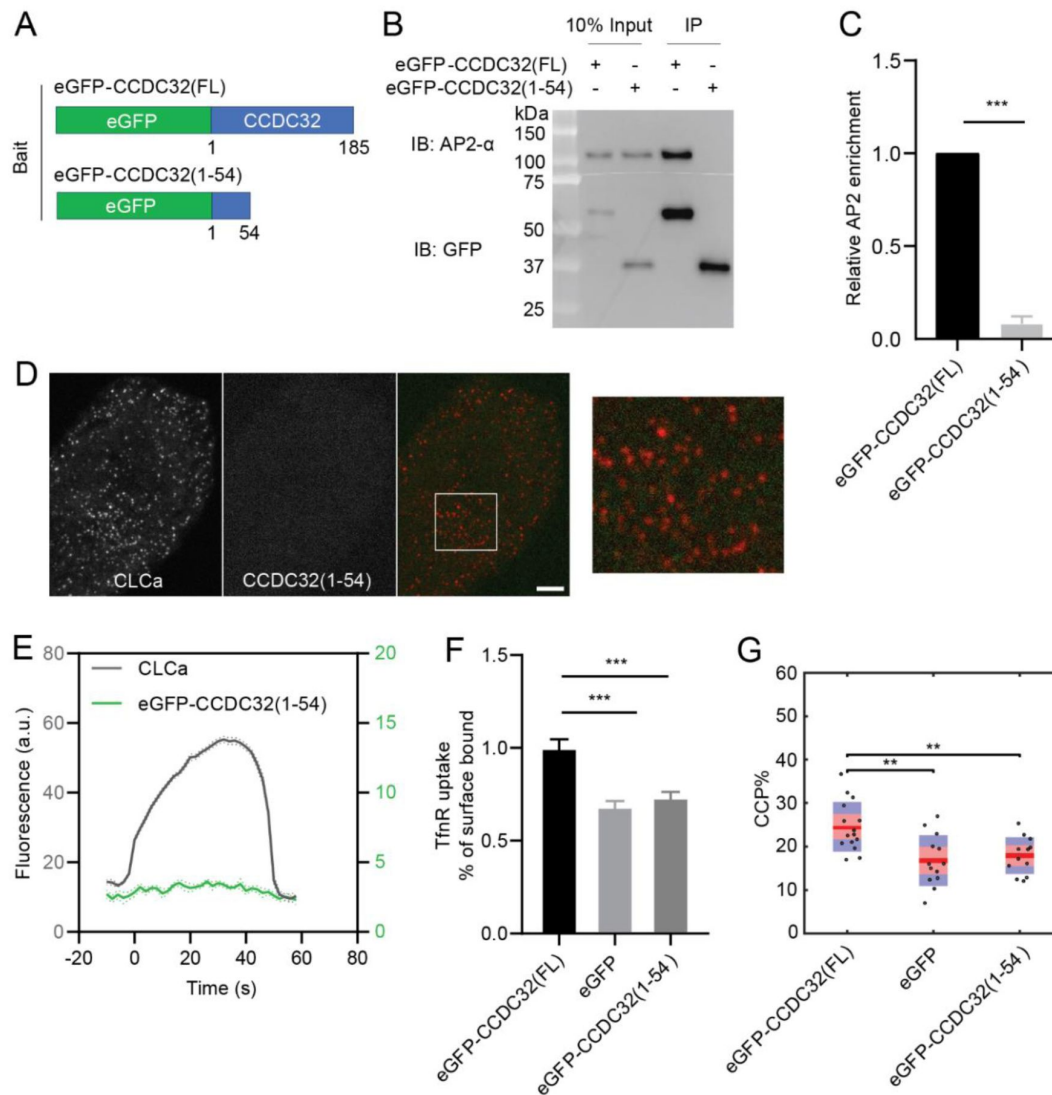


Figure 7

Disease-causing nonsense mutation in CCDC32 loses AP2 interaction capacity and inhibits CME.

(A-C) IP of eGFP from ARPE-HPV cells that stably express eGFP-CCDC32(FL) or eGFP-CCDC32(1-54) using anti-GFP magnetic beads. **(A)** The domain structures of eGFP-CCDC32(FL) and eGFP-CCDC32(1-54). **(B)** Representative immunoblotting result of n=3 IP samples. **(C)** Relative AP2 enrichments quantified from immunoblotting results. **(D)** Representative TIRFM images of ARPE-HPV cells that stably express mRuby-CLCa and eGFP-CCDC32(1-54). White ROI is magnified on the right. Scale bar = 5μm. **(E)** Cohort-averaged fluorescence intensity traces of CCPs and CCP-enriched eGFP-CCDC32(1-54). Number of tracks analyzed: 28658. **(F-G)** (F) TfR uptake efficiency and (G) CCP% of ARPE-HPV cells that stably express eGFP-CCDC32(FL), eGFP, or eGFP-CCDC32(1-54), and with siRNA-mediated knockdown of endogenous CCDC32. Each dot in (G) represents a movie. Statistical analysis of the data in (F) is two-tailed student's t-test: ***, $p \leq 0.001$. Statistical analysis of the data in (G) is the Wilcoxon rank sum test, **, $p \leq 0.01$.

fide CCPs and an accumulation of flat clathrin assemblies, unstable intermediates in CCP formation (Fig. 8). Despite these profound alterations in CCP dynamics, CME itself, as measured by TfnR internalization efficiency, is only partially inhibited. We speculate that this mild endocytic defect reflects the plasticity and resilience of CME. Indeed, we detect two potential compensatory mechanisms that occur upon depletion of CCDC32, namely an increase in the rate of CCS assembly and a decrease in the lifetimes of dynamic *bona fide* CCPs (i.e. an increased rate of CCP maturation) (Fig. 8).

While individual depletion of other endocytic accessory proteins has been reported to result in one or more of these phenotypes, to our knowledge, this combination of phenotypes has not been detected. Instead, either CCP assembly is reduced (Henne et al., 2010; Umasankar et al., 2014) or CCPs accumulate at later stages of invagination (McMahon and Boucrot, 2011; Messa et al., 2014). This combination of phenotypes has, however, been observed previously in cells expressing a truncated mutant of AP2 lacking the α -AD (Aguet et al., 2013). These cells also exhibited: i) near normal Tfn uptake, ii) an accumulation of flat clathrin assemblies, which are rapidly turned over, iii) an increase in the rate of nascent CCS initiation, iv) impaired stabilization of dynamic CCPs and, v) a decrease in productive CCP lifetimes. We speculate that loss of CCDC32 recruitment could have accounted for these unique phenotypes.

Although the mechanism of CCP invagination has been an important topic for decades, which endocytic accessory proteins (if any) regulate CCP invagination and (if so) how, has remained unclear. The depletion of CCDC32 in cells strongly inhibited the formation of dome-shaped and spherical CCSs, and instead resulted in the accumulation of flat clathrin assemblies. As the area of these clathrin lattices was insufficient to form a complete CCV, they likely represent defective CCP assembly intermediates that are rapidly turned over as abortive CCPs. This experimentally observed requirement of CCDC32 in CCP invagination is consistent with previous findings that clathrin assembly alone was not sufficient to induce CCP invagination in cells (Aguet et al., 2013), potentially due to the reported flexibility of clathrin coats (Ferguson et al., 2008; Tagiltsev et al., 2021). The mechanism by which CCDC32 promotes CCP invagination remains to be determined, but other domains of the protein may bind to and stabilize the clathrin coat, or the predicted largely disordered structure of CCDC32 (Fig. S6) may generate membrane curvature by molecular crowding (Busch et al., 2015).

We localized the AP2 binding site on CCDC32, which is essential for CCDC32-AP2 interactions *in vivo*, to a short, predicted alpha-helical region (aa78-98). Interestingly, this sequence, LASLEKKLRRIKGLNQEVTSKD, does not encode any of the known AP2 α -AD binding motifs (e.g. DxF/W, FxDxF, WxxF/W, FxxFxxL) commonly shared amongst other AP2 interacting partners (Olesen et al., 2008). However, we identified two canonical α -AD binding motifs in CCDC32 (¹⁷DLW¹⁹ and ³⁹FSDSF⁴³), both of which can be docked on the AP2 appendage domain with high confidence using AlphaFold 3.0 (Fig. S6D,E). While these binding motifs may have been sufficient for *in vitro* interactions with α -AD, they were not sufficient for AP2 binding *in vivo*. Unexpectedly, AlphaFold 3.0 modeling predicts, with high confidence, that the α -helical region we identified as essential for AP2 binding *in vivo* docks to an alpha helix formed by aa418-436 of the α -subunit (Fig. S6D,F), which is not encoded in our α -AD construct. This site is located in the α -trunk and buried with the AP2 core domain (Collins et al., 2002). However, it becomes accessible when the C-terminal domain of μ 2 adopts the open conformation triggered by cargo binding (Jackson et al., 2010). Thus, we speculate that CCDC32 may preferentially interact with membrane-bound AP2. Interestingly, while co-IP with CCDC32(Δ 78-99) no longer pulled down the intact AP2 complex or the α : σ 2 hemicomplex, it remained capable of pulling down the β 2: μ 2 hemicomplex. This result has a number of interesting implications, first that the extended full length CCDC32 has multiple points of interaction with AP2 and its subunits, and second, that the AP2 complex is less stable than previously assumed (i.e. the 2 hemicomplexes must be in flux). Further structural studies will be needed to identify the full extent of CCDC32 interactions with the AP2 complex.

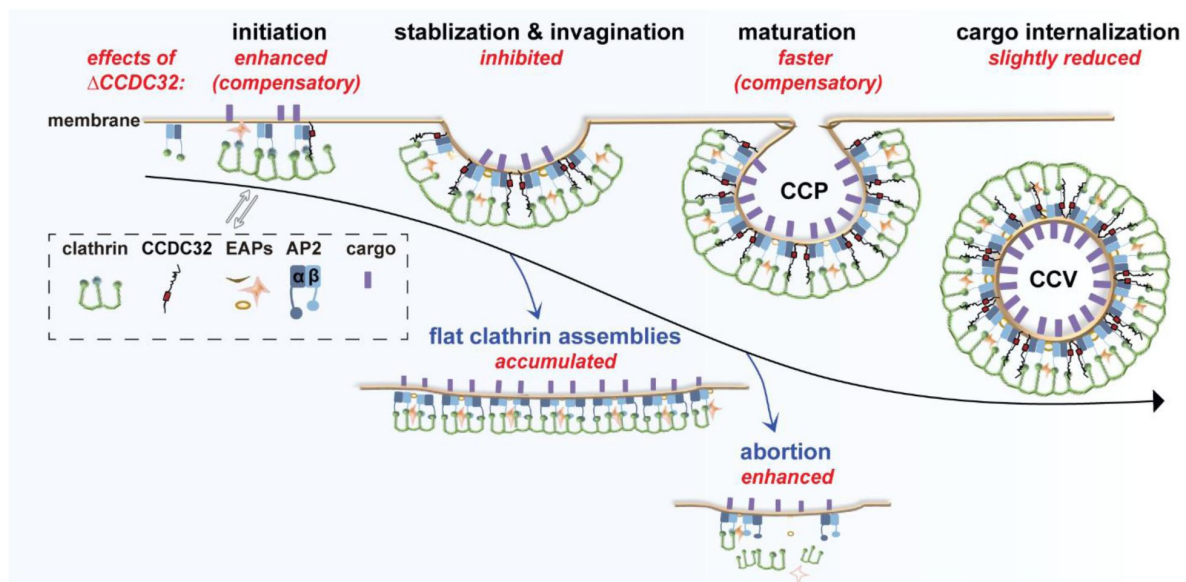


Figure 8

Cartoon illustration of CCDC32-AP2 interactions that regulate CME.

Interactions between residues 78-98, corresponding to a central α -helix of CCDC32, and AP2 are essential for recruitment of CCDC32 to CCPs. Depleting CCDC32 (Δ CCDC32) inhibits CCP stabilization and invagination, resulting in enhanced CCS abortion and accumulated flat clathrin assembly intermediates. As a compensation effect, CCS initiation is enhanced and CCP maturation is faster, resulting in only slightly reduced cargo internalization. EAPs: endocytic accessory proteins.

While this paper was under review, it was reported that CCDC32 functions together with AAGAB, as an essential chaperone for AP2 assembly (Wan et al., 2024 [\[1\]](#)). While there are discrepancies, our results are not incompatible with their findings. Thus, it is possible, even likely, that the ~40% residual CCDC32 present after siRNA knockdown may be sufficient to fulfill its catalytic chaperone activity in facilitating AP2 complex assembly, but not its presumed structural role in regulating early stages of CME. The inability of Wan et al. to detect CCDC32 binding to mature AP2 complexes or $\beta_2\mu_2$ hemicomplexes or its recruitment to CCPs may reflect a perturbed function of the C-terminally tagged construct (Wan et al., 2024 [\[1\]](#)). Indeed, their images show the formation of large (>100 nm), cytoplasmic puncta of CCDC32-GFP, which might be reflective of protein aggregation. Moreover, our data show a role for the C-terminus in diffuse plasma membrane binding. Finally, we find that the $\Delta 78-98$ mutant retains its ability to bind the $\beta_2\mu_2$ hemicomplex, but loses its ability to bind $\alpha:\sigma_2$ hemicomplex. Thus, the most parsimonious conclusion is that CCDC32 is multifunctional, acting catalytically to facilitate AP2 assembly and, after recruitment to CCPs, to regulate early stages of CME. Further studies are needed to fully elucidate the mechanisms governing each of these functions.

Our results provide a compelling explanation for how the clinically observed loss-of-function mutations in CCDC32 result in the development of CFNDS. As we have shown, the disease-associated CCDC32 loss-of-function mutants are not capable of interacting with intact AP2, thus losing their AP2 regulation capacity. Our results suggest that the inability to bind mature AP2 and hence to be recruited to nascent CCSs inhibits critical early stages of CME, and contributes to the development of CFNDS. Indeed, the clinical features of CCDC32 loss-of-function are similar to those resulting from AP2 loss-of-function mutations (Arrigo and Lin, 2021 [\[2\]](#); Gorvin et al., 2017 [\[3\]](#); Helbig et al., 2019 [\[4\]](#); Jung et al., 2015 [\[5\]](#); Li et al., 2018 [\[6\]](#); Li et al., 2010 [\[7\]](#)), supporting that CCDC32 functions through AP2. However, we suggest that the disease-associated CFNDS mutants are hypomorphic, as complete loss of AP2 complexes has been shown to be embryonic lethal in *Drosophila* (González-Gaitán and Jäckle, 1997 [\[8\]](#)) and zebrafish (Umasankar et al., 2012 [\[9\]](#)).

In summary, our study identified CCDC32 as an important endocytic accessory protein that regulates CCP stabilization and invagination via its interactions with AP2. Future work needs to address the multifunctional interactions of CCDC32 with AP2 and to define exactly how CCDC32 enhances curvature generation and coat stabilization.

Materials and methods

Plasmids

The eGFP-CCDC32(FL, human) cDNA in a pEGFP-C1 vector was purchased from Addgene (#110505) and then mutated to be siRNA resistant, which retained amino acid sequence (#98-102) while modifying the nucleotide sequence. Next, aa78-98 was deleted from siRNA resistant eGFP-CCDC32(FL) to generate eGFP-CCDC32($\Delta 78-98$). Finally, eGFP, eGFP-CCDC32(1-54), eGFP-CCDC32($\Delta 78-98$), and eGFP-CCDC32(FL) were separately cloned into a pLVx-CMV100 vector (Dean et al., 2016 [\[10\]](#)) using NEBuilder® HiFi DNA Assembly Master Mix (Catalog #E2621). A full list of primers used for mutagenesis and cloning is available in **Table S2** [\[11\]](#).

In addition, mRuby-CLCa in a pLVx-IRES-puro vector was generated in our previous study (Srinivasan et al., 2018 [\[12\]](#)). AP2- α -AD (aa701-938, mouse) in a pGEX-2T-1 vector and AP2- β -AD (aa592-937, rat) in a pGEX-4T-1 vector were kind gifts of the late Linton Traub (University of Pittsburgh, PA).

Cell culture, lentivirus infection, siRNA transfection and rescue

ARPE-19 and ARPE19-HPV16 (herein called ARPE-HPV) cells were obtained from ATCC and cultured in DMEM/F12 (Gibco, Catalog #8122502) with 10% FBS. HEK293T cells were obtained from ATCC and cultured in DMEM (Gibco, Catalog #8122070) with 10% FBS. ARPE-HPV cells that stably express eGFP-CLCa were generated in our previous study (Chen et al., 2020 [DOI](#)). ARPE-HPV cells that stably express a fully functional AP2- α -eGFP, in which eGFP is inserted into the flexible linker of AP2, at aa649, were generated in our previous study (Mino et al., 2020 [DOI](#)) and this AP2- α -eGFP construct has been shown to be fully functional.

Lentiviruses encoding mRuby-CLCa were produced in HEK293T packaging cells following standard transfection protocols (Kutner et al., 2009 [DOI](#)) and harvested for subsequent infections to ARPE-HPV cells to generate ARPE-HPV mRuby-CLCa cells. Lentiviruses encoding eGFP and siRNA-resistant, eGFP-CCDC32(1-54), eGFP-CCDC32(Δ 78-98), and eGFP-CCDC32(FL) were produced in HEK293T packaging cells and harvested for subsequent infections to ARPE-HPV mRuby-CLCa cells. Infected cells were FACS sorted for homogenous mRuby and eGFP signals after 3 days and passaged for 2 weeks before experiments.

For siRNA-mediated knockdown of CCDC32, ARPE-HPV cells stably expressing eGFP-CLCa were seeded on 6-well plates (250,000 cells/well) and transfected with 2 rounds of siCCDC32 (Silencer Select Pre-designed siRNA ID#: s228444, targets aa98-102 sequence) through 3 days. Cells treated with siControl (Silencer Select Negative Control #1 siRNA, cat#:4390843) were used as negative control. Transfections of siRNA were mediated with Opti-MEM and Lipofectamine RNAi-MAX (Invitrogen) as detailed in ref. (Chen et al., 2020 [DOI](#)).

For CCDC32 rescue experiments, ARPE-HPV cells stably expressing eGFP or siRNA-resistant eGFP-CCDC32(FL) eGFP, eGFP-CCDC32(Δ 78-98 or eGFP-CCDC32(1-54) were transfected with siCCDC32 or siControl as described above. Anti-GFP Monoclonal Antibody (Proteintech, # 66002-1-Ig), Anti-Vinculin Polyclonal Antibody (Proteintech, #26520-1-AP), and anti-C15orf57 Polyclonal antibody (Invitrogen, #PA5-98982) were used in Western Blotting to confirm protein expression level and knockdown efficiency.

Co-immunoprecipitation (co-IP)

ARPE-HPV cells with stable expression of AP2- α -eGFP, eGFP, eGFP-CCDC32(1-54), eGFP-CCDC32(Δ 78-98) or eGFP-CCDC32(FL) were cultured in a 15-cm dish. When confluence reached ~90%, cells were washed 3x with ice-cold PBS and detached with cell scratcher. Next, cells were spun down at 500 g, 4°C for 3 min, and then resuspended in 1ml ice-cold lysis buffer (50 mM Tris, pH 7.5, 150 mM NaCl, 1mM EDTA, 0.5% Triton X-100, 1 $\times$ protease inhibitor, 1mM PMSF). After 30 min rotation in a cold room and occasional vortexing, the lysed cells were spun at 500 g, 4°C, for 3 min to remove the nuclei. Subsequently, 50 μ l anti-GFP magnetic beads (Biolinkedin, #L1016) were added to cell lysate that contained 0.5mg proteins (determined by BCA assay). The reaction was allowed to proceed by rotation at 4°C for 2h and then precipitated with DynaMag. The sediments were washed with lysis buffer, and then resuspended and heat-denatured in 2x Laemmli buffer (supplemented with 5% β -Mercaptoethanol). The final samples were run into SDS-PAGE gels before: 1) being transferred to membranes for Western Blotting; or 2) being sent to the proteomics core facility for mass spectrometry analysis. Anti- α -Adaptin 1/2 Antibody (C-8) (Santa Cruz Biotechnology, Catalog #sc-17771) was used in WB to detect AP2 enrichments.

Mass Spectrometry Analysis

Samples were digested overnight with trypsin (Pierce) following reduction and alkylation with DTT and iodoacetamide (Sigma-Aldrich). Following solid-phase extraction cleanup with an Oasis HLB μ elution plate (Waters), the resulting peptides were reconstituted in 10 μ L of 2% (v/v)

acetonitrile (ACN) and 0.1% trifluoroacetic acid in water. 2 μ L of each sample were injected onto an Orbitrap Fusion Lumos mass spectrometer (Thermo Electron) coupled to an Ultimate 3000 RSLC-Nano liquid chromatography systems (Dionex). Samples were injected onto a 75 μ m i.d., 75-cm long EasySpray column (Thermo), and eluted with a gradient from 0-28% buffer B over 90 min. Buffer A contained 2% (v/v) ACN and 0.1% formic acid in water, and buffer B contained 80% (v/v) ACN, 10% (v/v) trifluoroethanol, and 0.1% formic acid in water. The mass spectrometer operated in positive ion mode with a source voltage of 2.4 kV and an ion transfer tube temperature of 275 $^{\circ}$ C. MS scans were acquired at 120,000 resolution in the Orbitrap and up to 10 MS/MS spectra were obtained in the Orbitrap for each full spectrum acquired using higher-energy collisional dissociation (HCD) for ions with charges 2-7. Dynamic exclusion was set for 25 s after an ion was selected for fragmentation.

Raw MS data files were analyzed using Proteome Discoverer v2.4 (Thermo), with peptide identification performed using Sequest HT searching against the human reviewed protein database from UniProt. Fragment and precursor tolerances of 10 ppm and 0.6 Da were specified, and three missed cleavages were allowed. Carbamidomethylation of Cys was set as a fixed modification and oxidation of Met was set as a variable modification. The false-discovery rate (FDR) cutoff was 1% for all peptides. To generate the volcano plot, Tubulin Beta 6 was used for Normalization.

The data was uploaded to the MassIVE data repository with accession number MSV000095338.

Protein purification and GST pull-down assay

GST in a pGEX-6P-1 vector, GST-AP2- α -AD in a pGEX-2T-1 vector, and GST-AP2- β -AD in a pGEX-4T-1 vector were transfected and expressed in BL21(DE3) separately, and then affinity purified using GSTrap HP column (Cytiva). The affinity-purified GST fusion proteins were applied to a HiLoad 16/600 Superdex 200 pg column (Cytiva). Target proteins were collected and concentrated using Amicon Ultra-15 10K Centrifugal filters (Sigma-Aldrich), and stored in 20 mM HEPES, 150 mM NaCl, 1mM TCEP, pH 7.4.

In GST pull-down assays, purified GST, GST-AP2- α -AD, and GST-AP2- β -AD were pre-bound to anti-GST beads (Biolinkedin, #L-2004). Subsequently, these beads were separately added into 0.5 mg cell lysate of ARPE-HPV eGFP-CCDC32(FL). The reactions were allowed to proceed by rotation at 4 $^{\circ}$ C for 2h, and then the beads were spun down at 1000 g, 4 $^{\circ}$ C for 3 min. The sediments were washed with lysis buffer, and then heat-denatured in 2x Laemmli buffer (supplemented with 5% β -Mercaptoethanol) before running into SDS-PAGE gels, which followed by Western Blotting analysis.

Transferrin receptor (TfnR) uptake assay

Internalization of TfnR was quantified by in cell ELISA following established protocols described (Chen et al., 2020 [\[1\]](#); Conner and Schmid, 2003 [\[2\]](#); Reis et al., 2015 [\[3\]](#); Srinivasan et al., 2018 [\[4\]](#)). Briefly, ~15,000 cells were seeded on gelatin-coated 96-well plate overnight. Next day, cells were starved in 37 $^{\circ}$ C-warm PBS⁴⁺ (1 $\times$ PBS buffer plus 0.2% BSA, 1mM CaCl₂, 1mM MgCl₂, and 5mM D-glucose) for 30 min. After starvation, the cell medium was replaced with ice-cold PBS⁴⁺ containing 5 μ g/ml HTR-D65 (anti-TfnR mAb) (Schmid and Smythe, 1991 [\[5\]](#)). In parallel, some cells were kept at 4 $^{\circ}$ C for the measurement cell surface TfnR (denoted as S) and blank controls (denoted as B), while some were incubated in a 37 $^{\circ}$ C water bath for the indicated times for the measurement of internalized TfnR (denoted as I). For B and I, acid wash (0.2 M acetic acid and 0.2 M NaCl, pH 2.3) was applied to remove surface-bound HTR-D65. After washing with cold PBS. All cells were fixed with 4% PFA (Electron Microscopy Sciences) diluted in PBS for 30 min at 37 $^{\circ}$ C. Subsequently, the cells were permeabilized with 0.1% Triton X-100 and blocked with Q-PBS (PBS, 2% BSA, 0.1%

lysine, and 0.01% saponin, pH 7.4) for 2h. Surface-bound and internalized HTR-D65 were detected with HRP Goat anti-Mouse IgG (H+L) (BioRad) and o-phenylenediamine dihydrochloride (OPD, Sigma-Aldrich). Well-to-well variation of cell numbers was accounted for by BCA assays.

TIRF and Epi-TIRF microscopy

Cells for imaging were seeded on gelatin-coated glass bottom dishes 35 mm (ibidi, #81218-800) for ~12h before data acquisition. Live cell imaging was conducted with a Nikon Eclipse Ti2 inverted microscope that was equipped with: 1) an Apo TIRF/100x 1.49 Oil objective; 2) a Prime Back Illuminated sCMOS Camera (Prime BSI, 6.5 x 6.5µm pixel size and 95% peak quantum efficiency; 3) a M-TIRF module for epifluorescence (Epi) acquisition; 4) an H-TIRF module for TIRF acquisition, where penetration depth was fixed to 80nm; 5) an Okolab Cage Incubator for maintaining 37°C and 5% CO₂.

For time-lapse TIRF imaging, 451 consecutive images were acquired at a frame rate of 1 frame/s for single channel or 0.5 frame/s for dual channels. For time-lapse Epi-TIRF imaging, 451 consecutive Epi and TIRF images were acquired nearly simultaneously at a frame rate of 0.66 frame/s. Perfect Focus System (PFS) was applied during time-lapse imaging.

The acquired data was analyzed using cmeAnalysis (Aquet et al., 2013 [DOI](#); Jaqaman et al., 2008 [DOI](#)) and DASC (Wang et al., 2020 [DOI](#)), which are available at: <https://github.com/DanuserLab/cmeAnalysis> [DOI](#). Detailed protocols are available via this link. Briefly, cmeAnalysis was used to track the lifetime and fluorescence of clathrin-coated structures, and then DASC was applied to unbiasedly classify *bona fide* CCPs vs. abortive coats. Next, the classified CCP tracks were used to calculate CCP invagination (Δz), as previously described (Wang et al., 2020 [DOI](#)). Tracks that overlap with others or deviate from the properties of a diffraction-limited particle were excluded from the analysis.

Platinum Replica Electron Microscopy (PREM)

The “unroofing” technique that mechanically removes the upper cell plasma membrane while preserving the ventral plasma membrane with CCSs and other structures for PREM analyses was performed essentially as described previously (Yang et al., 2022 [DOI](#)). Briefly, ARPE19 cells treated with siCCDC32 or siControl on coverslips were quickly transferred into ice-cold PEM buffer (100 mM PIPES-KOH, pH 6.9, 1 mM MgCl₂ and 1 mM EGTA) containing 2 µM unlabeled phalloidin (Sigma, #P2141) and 10 µM taxol (Sigma-Aldrich, #T7402) and unroofed by a brief (1s) ultrasonic burst from a 1/8-inch microprobe positioned at ~45° angle ~3 mm above the coverslip and operated by Misonix XL2020 Ultrasonic Processor at 17-20% of output power. After sonication, the coverslips were immediately fixed with 2% glutaraldehyde in 0.1 M Na-cacodylate buffer, pH 7.3 for at least 20 min at room temperature.

Sample processing for PREM was performed as described previously (Svitkina, 2022 [DOI](#)). In brief, glutaraldehyde-fixed cells were post-fixed by sequential treatment with 0.1% tannic acid and 0.2% uranyl acetate in water, critical-point dried, coated with platinum and carbon, and transferred onto EM grids for observation.

PREM samples were examined using JEM 1011 transmission electron microscope (JEOL USA, Peabody, MA) operated at 100 kV. Images were acquired by an ORIUS 832.10W CCD camera (Gatan, Warrendale, PA) and presented in inverted contrast.

Based on the degree of invagination, the shapes of CCSs were classified into: flat CCSs with no obvious invagination; dome-shaped CCSs that had a hemispherical or less invaginated shape with visible edges of the clathrin lattice; and spherical CCSs that had a round shape with the invisible edges of clathrin lattice in 2D projection images. In most cases, the shapes were obvious in 2D

PREM images. In uncertain cases, the degree of CCS invagination was determined using images tilted at ± 10 – 20 degrees. The area of CCSs were measured using ImageJ and used for the calculation of the CCS occupancy on the plasma membrane.

Online supplemental material

Fig. S1 [↗](#) shows the western blotting results that indicate the expression levels of exogenous CCDC32. **Fig. S2** [↗](#) shows the representative TIRFM images of ARPE-HPV cells that stably express mRuby-CLCa and eGFP. It also shows the cohort-averaged fluorescence intensity traces of CCPs and CCP-enriched eGFP. Videos 1 and 2 shows addition time-lapse TIRFM imaging data for **Fig. 2** [↗](#). Videos are accelerated 25-fold. **Fig. S3** [↗](#) shows that siRNA-mediated knockdown of CCDC32 did not affect AP2 protein levels. **Fig. S4** [↗](#) shows the fractions of CCPs in lifetime cohorts. **Fig. S5** [↗](#) shows additional data for **Fig. 4** [↗](#). **Fig. S6** [↗](#) shows the AlphaFold 3.0 prediction of CCDC32 monomer structure and CCDC32-AP2- α complex structure. **Fig. S7** [↗](#) shows a collection of representative TIRFM images from the main text showing cells that stably express the same amount of eGFP-CCDC32(FL), eGFP, eGFP-CCDC32($\Delta 78$ -98), and eGFP-CCDC32(1-54). **Table S1** [↗](#) shows additional data for **Fig. 5** [↗](#). **Table S2** [↗](#) shows a list of primers used in this study.

Data availability

Original data are available from the corresponding authors upon request.

Supplementary Information

Figure S1

(A) Western blotting indicates similar expression levels of exogenous eGFP-CCDC32(FL), eGFP, eGFP-CCDC32(1-54) and eGFP-CCDC32(Δ 78-98) in ARPE-HPV mRuby-CLCa cells. Note that the anti-CCDC32 antibody does not detect the eGFP-CCDC32(Δ 78-98) as well as full-length and is unable to detect eGFP-CCDC32(1-54); **(B)** siCCDC32 knocked down endogenous CCDC32 but not exogenously introduced eGFP-CCDC32(FL) from ARPE-HPV mRuby-CLCa cells.

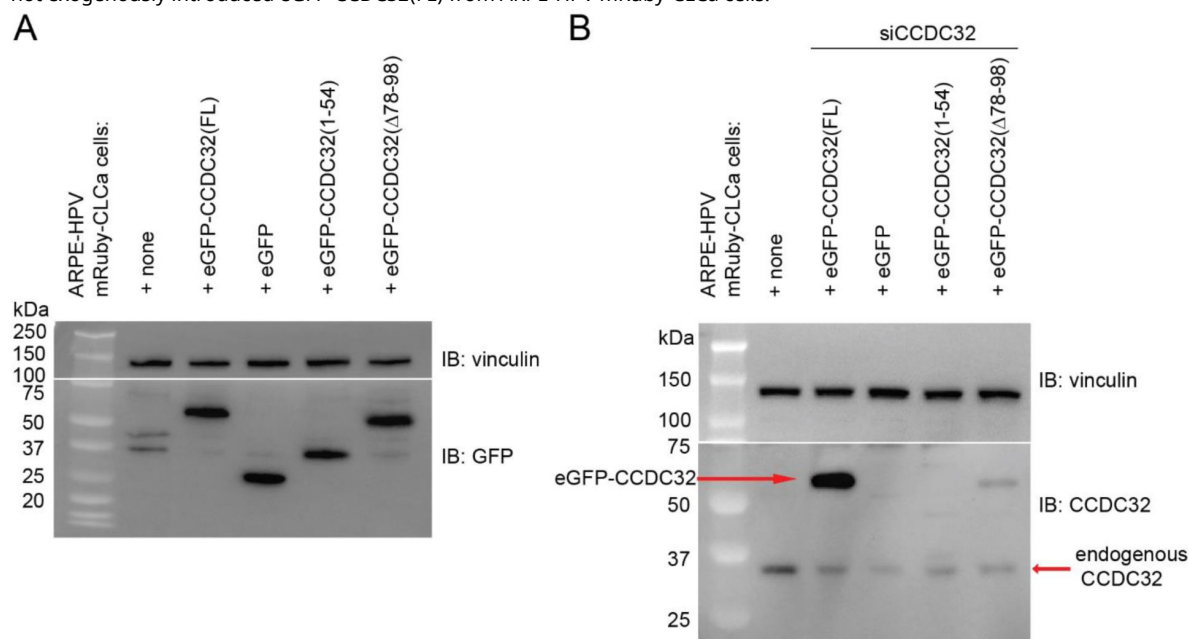


Figure S2

(A) Representative TIRFM images of ARPE-HPV cells that stably express mRuby-CLCa and EGFP. White ROI is magnified on the right. Scale bar = 5 μ m. **(B)** Cohort-averaged fluorescence intensity traces of CCPs and CCP-enriched EGFP. Number of tracks analyzed: 46833.

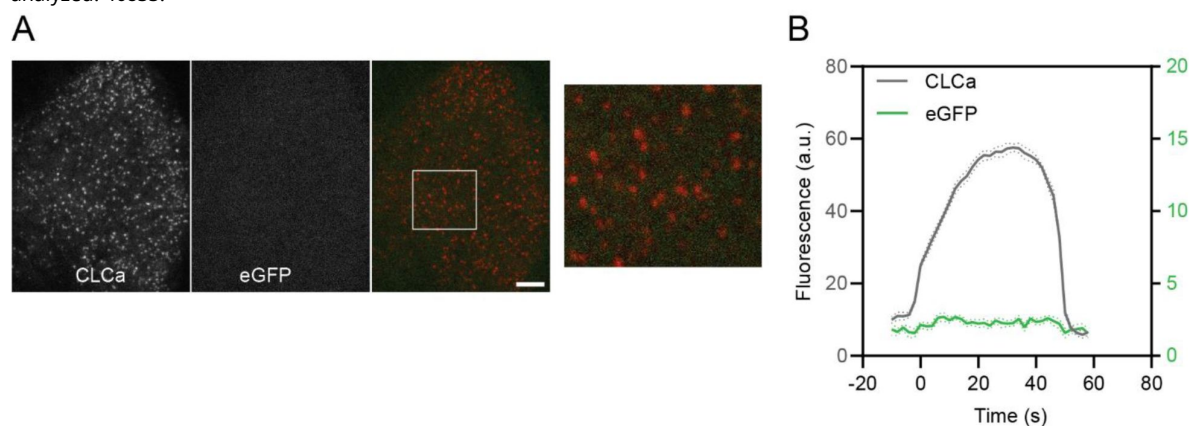


Figure S3

(A-B) siRNA-mediated knockdown of CCDC32 does not affect AP2 expression level. **(A)** Representative immunoblotting result of AP2 subunits from n=3 biological repeats. **(B)** Quantification of results from (A). Error bars indicate standard deviations. Statistical analysis is student's t-test: ns, not significant. **(C)** All the four subunits of intact AP2 efficiently co-IP with eGFP-CCDC32(FL).

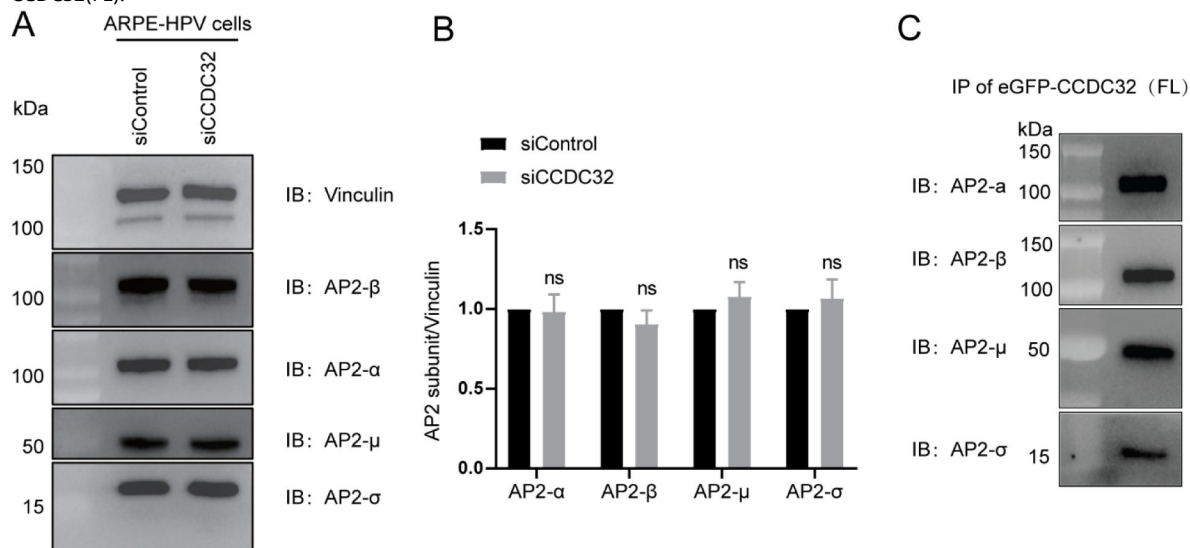
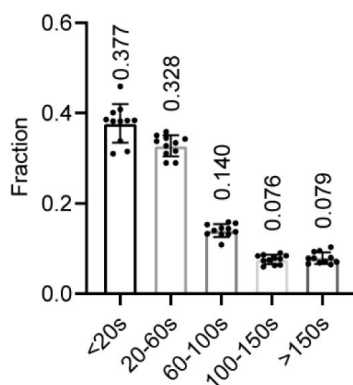


Figure S4

Fraction of CCPs in lifetime cohorts. Data was obtained from ARPE-HPV eGFP-CLCa cells that were treated with siCCDC32. The obtained data was processed with cmeAnalysis.



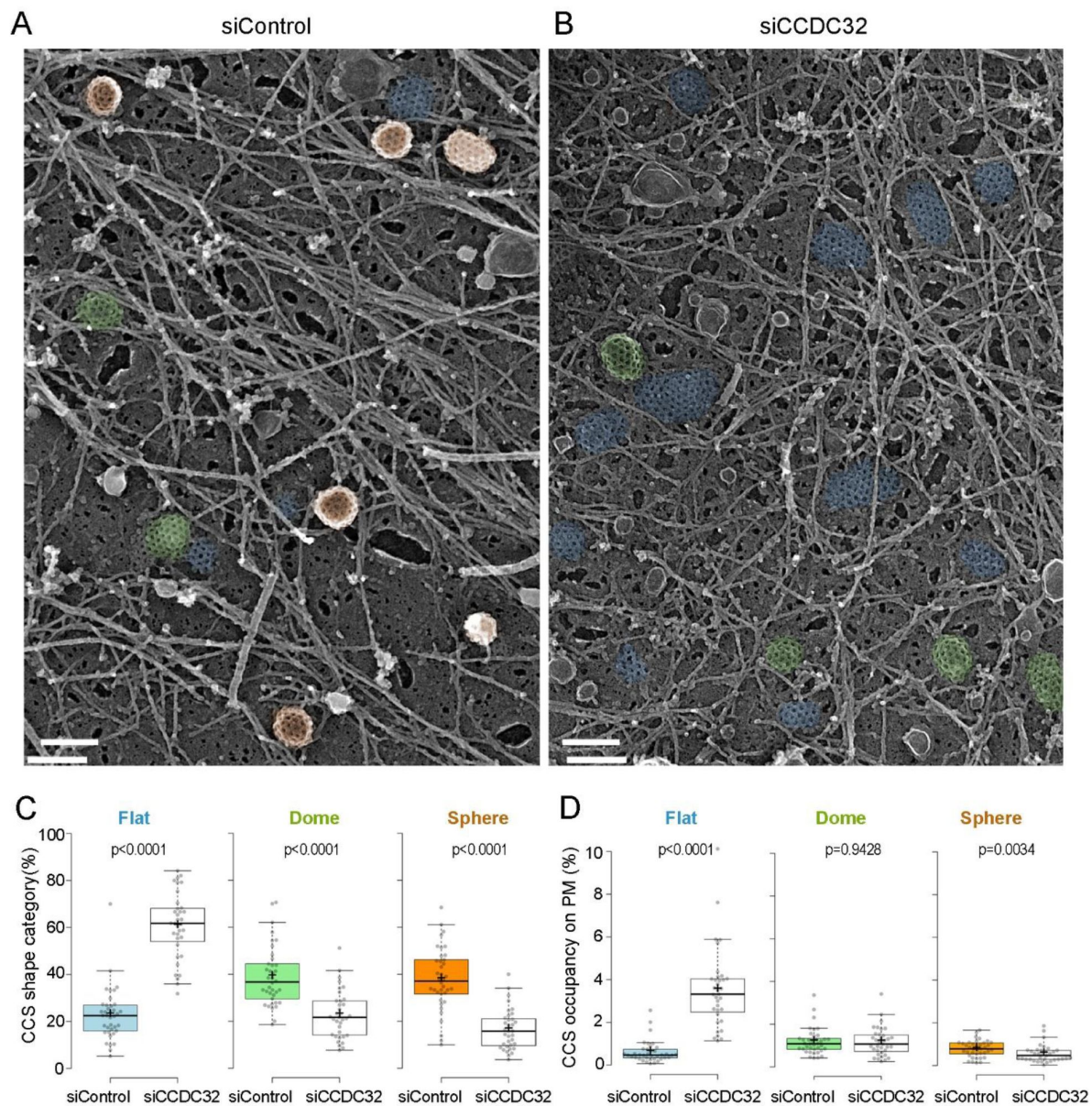


Figure S5

CDC32 knockdown shifts the shapes of clathrin-coated structures.

(A-B) Representative PREM images of ARPE-HPV cells treated with control siRNA (A) or CDC32 siRNA (B). Scale bars = 200 nm. (C-D) Quantification of the CCS shape category (C) and CCS occupancy on PM (D). Each dot represents an individual fragment of the plasma membrane; the number of plasma membrane fragments analyzed is 38 for siControl and 35 for siCDC32 from two independent experiments. Statistical tests were performed using either unpaired t-test or Mann-Whitney test depending on normality of distributed values.

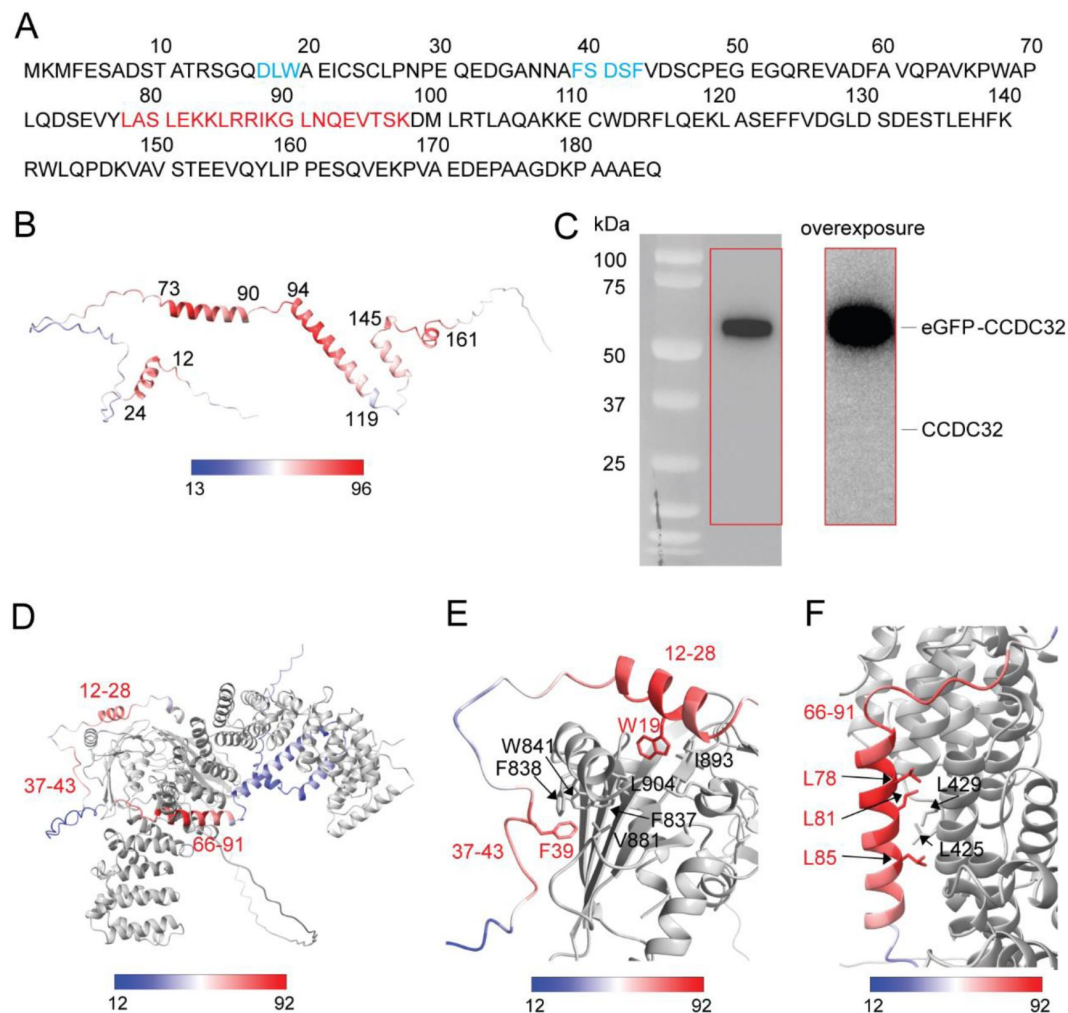


Figure S6

The sequence and structure of CCDC32.

(A) Amino acid sequence of CCDC32. Blue font: canonical AP2 binding motifs; red font: predicted coiled-coil domain (aa78-98). (B) AlphaFold 3.0 modelling of full-length CCDC32 (Jumper et al., 2021 [\[1\]](#); Varadi et al., 2021 [\[2\]](#)). Residues with pLDDT > 0.8 are labelled in red. (C) co-IP of eGFP-CCDC32 did not pulldown endogenous CCDC32. Red-cycled area was overexposed on the right. (D-E) AlphaFold 3.0 modelling of CCDC32 interaction with AP2-α (gene: AP2A2). For all panels, CCDC32 is colored by pLDDT and AP2-α is grey. (D) CCDC32-AP2A2 complex. CCDC32 residues with pLDDT > 0.7 are labelled in red. (E-F) Close up view of CCDC32-AP2A2 complex near CCDC32 residues (E) 12-28, 37-43 and (F) 66-91. Hydrophobic CCDC32 and AP2-α residues at the interface are labelled in red and black, respectively. Two canonical α-AD binding motifs in CCDC32 (¹⁷DLW¹⁹ and ³⁹FSDS⁴³) can be docked on the AP2 appendage domain with high confidence; aa78-98 docks with high confidence to an alpha helix in the α-subunit encoded by aa418-438. AlphaFold 3.0 modelling was performed using AlphaFold Server (Abramson et al., 2024 [\[3\]](#)) and structure images were generated using ChimeraX (Meng et al., 2023 [\[4\]](#)).

Figure S7

A collection of representative TIRFM images from the main text showing cells that stably express the same amount of eGFP-CCDC32(FL), eGFP, eGFP-CCDC32(Δ 78-98), and eGFP-CCDC32(1-54).

Diffuse PM fluorescence is greatest for eGFP-CCDC32(FL), diminished, but detectable for eGFP-CCDC32(Δ 78-98), but absent for eGFP-CCDC32(1-54) and control eGFP. Scale bar = 5 μ m. Blue dashed line indicates the background PM fluorescence of eGFP.

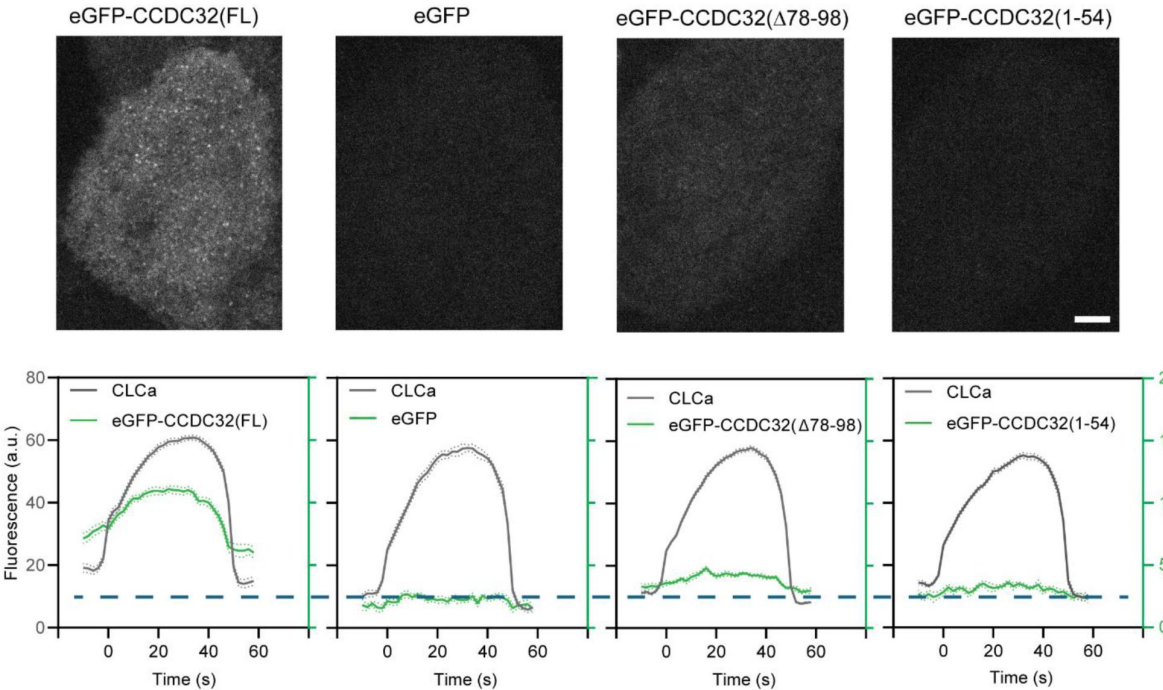


Table S1

ID and abundances of key proteins detected in IP samples by Mass Spectroscopy analysis.

#1-#3: 3 independent repeats of control IPs; #4-#6: 3 independent repeats of AP2- α -eGFP IPs.

	Abundance					
	Control IPs			AP2- α -eGFP IPs		
Gene	#1	#2	#3	#4	#5	#6
AP2A1	2.E+06	1.E+06	6.E+05	1.E+08	1.E+08	2.E+08
AP2A2	4.E+06	1.E+06	7.E+05	3.E+08	3.E+08	3.E+08
AP2B1	7.E+06	6.E+06	3.E+06	7.E+08	6.E+08	6.E+08
AP2M1	1.E+06	8.E+05	2.E+05	2.E+08	1.E+08	2.E+08
AP2S1	1.E+05	5.E+04	9.E+03	2.E+07	2.E+07	5.E+07
CCDC32	0.E+00	0.E+00	0.E+00	3.E+06	2.E+06	2.E+06
AAK1	7.E+04	1.E+05	3.E+04	2.E+07	1.E+07	2.E+07
EPS15	2.E+06	3.E+05	2.E+05	3.E+07	3.E+07	3.E+07
EPS15L	3.E+05	1.E+04	2.E+04	1.E+07	1.E+07	1.E+07
FCHO2	4.E+04	3.E+04	2.E+04	1.E+06	1.E+06	1.E+06
ITSN1	1.E+04	0.E+00	0.E+00	1.E+06	9.E+05	7.E+05
NECAP1	9.E+04	8.E+04	3.E+04	9.E+06	7.E+06	1.E+07
NECAP2	1.E+05	7.E+03	1.E+04	2.E+07	2.E+07	2.E+07
AAGAB	9.E+00	0.E+00	0.E+00	3.E+06	4.E+06	7.E+06

Mutagenesis	Forward primer	Reverse primer
eGFP- CCDC32(Δ 78-98)	GCAGGATTCAGAAGTGTATGAC ATGCTTCGAACCTCTG	CAGAGTTCGAAGCATGTCATAC ACTTCTGAATCCTGC
eGFP- CCDC32(FL)_siRe sistant	CAGGAAGTGACTTCCAAGGATA TGTACGCACCCTGGCCCAAGC CAAGAAGGA	TCCTTCTTGGCTTGGGCCAGGG TGCGTAACATATCCTTGGAAGTC ACTTCCTG
eGFP- CCDC32(Δ 78- 98)_siResistant	TCCCTTGCAGGATTCAGAAGTG TATGATATGTTACGCACCCTGGC CCAAGCCAAGAA	TTCTTGGCTTGGGCCAGGGTG CGTAACATATCATACTTCTGA ATCCTGCAAGGGA
Cloning	Forward primer	Reverse primer
backbone: pLVx- CMV100	CGCCGCTAGCGCTAACTAGTAT GGTGAGCAAGGGCGAG	GTCGGGCCCTCTAGACTCGAGT TACTGTTCTGCTGCTGCTG
insert gene: eGFP- CCDC32(FL)		
backbone: pLVx- CMV100	CGCCGCTAGCGCTAACTAGTAT GGTGAGCAAGGGCGAG	CGGGCCCTCTAGACTCGAGTTA CTTGACAGCTCGTCCATGC
insert gene: eGFP		
backbone: pLVx- CMV100	CGCCGCTAGCGCTAACTAGTAT GGTGAGCAAGGGCGAG	TCGGGCCCTCTAGACTCGAGCT ACCTCTGGCCTTCACCTTC
insert gene: eGFP-CCDC32(1- 54)		
backbone: pLVx- CMV100	CGCCGCTAGCGCTAACTAGTAT GGTGAGCAAGGGCGAG	GTCGGGCCCTCTAGACTCGAGT TACTGTTCTGCTGCTGCTG
insert gene: eGFP- CCDC32(Δ 78-98)		

Table S2

List of primers.

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Additional information

Author contributions

Z. Chen, Z. Yang, and C. Yang designed the experiments. Z. Chen, Z. Yang, C. Yang, T. Svitkina and S.L. Schmid interpreted the results and wrote the manuscript with input from all authors. Z. Yang, Y. Li and Z. Huang performed the TIRF and Epi-TIRF microscope imaging and data analysis. Z. Yang performed the co-IP, GST pull-down assays and TfnR uptake assays. C. Yang performed the PREM experiments. P. Xu performed FACS sorting. L. Han drew the cartoon illustrations. J. Pak did the AlphaFold3 modeling.

Additional files

Video 1. Time-lapse TIRFM imaging of ARPE-HPV eGFP-CLCa cells that were treated with control siRNA. Images were obtained at 1 frame/s and collected for 7.5 min. Video is accelerated 25-fold. [🔗](#)

Video 2. Time-lapse TIRFM imaging of ARPE-HPV eGFP-CLCa cells that were treated with CCDC32 siRNA. Images were obtained at 1 frame/s and collected for 7.5 min. Video is accelerated 25-fold. [🔗](#)

References

- Abdalla E., Alawi M., Meinecke P., Kutsche K., Harms F.L. (2022) **Cardiofacioneurodevelopmental syndrome: Report of a novel patient and expansion of the phenotype** *Am J Med Genet A* **188**:2448–2453 [Google Scholar](#)
- Abramson J., Adler J., Dunger J., Evans R., Green T., Pritzel A., Ronneberger O., Willmore L., Ballard A.J., Bambrick J., Bodenstein S.W., Evans D.A., Hung C.-C., O'Neill M., Reiman D., Tunyasuvunakool K., Wu Z., Žemgulytė A., Arvaniti E., Beattie C., Bertolli O., Bridgland A., Cherepanov A., Congreve M., Cowen-Rivers A.I., Cowie A., Figurnov M., Fuchs F.B., Gladman H., Jain R., Khan Y.A., Low C.M.R., Perlin K., Potapenko A., Savy P., Singh S., Stecula A., Thillaisundaram A., Tong C., Yakneen S., Zhong E.D., Zielinski M., Židek A., Bapst V., Kohli P., Jaderberg M., Hassabis D., Jumper J.M. (2024) **Accurate structure prediction of biomolecular interactions with AlphaFold 3** *Nature* **630**:493–500 [Google Scholar](#)
- Aguet F., Antonescu Costin N., Mettlen M., Schmid Sandra L., Danuser G. (2013) **Advances in Analysis of Low Signal-to-Noise Images Link Dynamin and AP2 to the Functions of an Endocytic Checkpoint** *Developmental Cell* **26**:279–291 [Google Scholar](#)
- Arrigo A.B., Lin J.I. (2021) **Endocytic Protein Defects in the Neural Crest Cell Lineage and Its Pathway Are Associated with Congenital Heart Defects** *Int J Mol Sci* **22** [Google Scholar](#)
- Baschieri F., Porshneva K., Montagnac G. (2020) **Frustrated clathrin-mediated endocytosis – causes and possible functions** *Journal of Cell Science* **133** [Google Scholar](#)
- Bhave M., Mino R.E., Wang X., Lee J., Grossman H.M., Lakoduk A.M., Danuser G., Schmid S.L., Mettlen M. (2020) **Functional characterization of 67 endocytic accessory proteins using multiparametric quantitative analysis of CCP dynamics** *Proc Natl Acad Sci U S A* **117**:31591–31602 [Google Scholar](#)
- Blue R.E., Curry E.G., Engels N.M., Lee E.Y., Giudice J. (2018) **How alternative splicing affects membrane-trafficking dynamics** *Journal of Cell Science* **131**:jcs216465 [Google Scholar](#)
- Busch D.J., Houser J.R., Hayden C.C., Sherman M.B., Lafer E.M., Stachowiak J.C. (2015) **Intrinsically disordered proteins drive membrane curvature** *Nat Commun* **6**:7875 [Google Scholar](#)
- Chen Z., Mino R.E., Mettlen M., Michaely P., Bhave M., Reed D.K., Schmid S.L. (2020) **Wbox2: A clathrin terminal domain-derived peptide inhibitor of clathrin-mediated endocytosis** *Journal of Cell Biology* **219** [Google Scholar](#)
- Chen Z., Schmid S.L. (2020) **Evolving models for assembling and shaping clathrin-coated pits** *Journal of Cell Biology* **219** [Google Scholar](#)
- Cho N.H., Cheveralls K.C., Brunner A.D., Kim K., Michaelis A.C., Raghavan P., Kobayashi H., Savy L., Li J.Y., Canaj H., Kim J.Y.S., Stewart E.M., Gnann C., McCarthy F., Cabrera J.P., Brunetti R.M., Chhun B.B., Dingle G., Hein M.Y., Huang B., Mehta S.B., Weissman J.S., Gómez-Sjöberg R., Itzhak

- D.N., Royer L.A., Mann M., Leonetti M.D. (2022) **OpenCell: Endogenous tagging for the cartography of human cellular organization** *Science* **375**:eabi6983 [Google Scholar](#)
- Collins B.M., McCoy A.J., Kent H.M., Evans P.R., Owen D.J. (2002) **Molecular architecture and functional model of the endocytic AP2 complex** *Cell* **109**:523–535 [Google Scholar](#)
- Conner S.D., Schmid S.L. (2003) **Differential requirements for AP-2 in clathrin-mediated endocytosis** *J Cell Biol* **162**:773–779 [Google Scholar](#)
- Dean K.M., Roudot P., Reis C.R., Welf E.S., Mettlen M., Fiolka R. (2016) **Diagonally Scanned Light-Sheet Microscopy for Fast Volumetric Imaging of Adherent Cells** *Biophys J* **110**:1456–1465 [Google Scholar](#)
- DeMari J., Mroske C., Tang S., Nimeh J., Miller R., Lebel R.R. (2016) **CLTC as a clinically novel gene associated with multiple malformations and developmental delay** *American Journal of Medical Genetics Part A* **170**:958–966 [Google Scholar](#)
- Elkin S.R., Bendris N., Reis C.R., Zhou Y., Xie Y., Huffman K.E., Minna J.D., Schmid S.L. (2015) **A Systematic Analysis Reveals Heterogeneous Changes in the Endocytic Activities of Cancer Cells** *Cancer Research* **75**:4640–4650 [Google Scholar](#)
- Ferguson M.L., Prasad K., Boukari H., Sackett D.L., Krueger S., Lafer E.M., Nossal R. (2008) **Clathrin triskelia show evidence of molecular flexibility** *Biophys J* **95**:1945–1955 [Google Scholar](#)
- Gilles Moulay J.L., Lemaître Mégane, Bitoun Marc, Furling Denis, Vassilopoulos Stéphane (2019) **Revealing alternative splicing control of clathrin structural plasticity in health and disease** In: *EMBO Conference Abstract* [Google Scholar](#)
- González-Gaitán M., Jäckle H. (1997) **Role of Drosophila alpha-adaptin in presynaptic vesicle recycling** *Cell* **88**:767–776 [Google Scholar](#)
- Gorvin C.M., Rogers A., Stewart M., Paudyal A., Hough T.A., Teboul L., Wells S., Brown S.D., Cox R.D., Thakker R.V. (2017) **N-ethyl-N-nitrosourea-Induced Adaptor Protein 2 Sigma Subunit 1 (Ap2s1) Mutations Establish Ap2s1 Loss-of-Function Mice** *JBMR Plus* **1**:3–15 [Google Scholar](#)
- Grove J., Metcalf D.J., Knight A.E., Wavre-Shapton S.T., Sun T., Protonotarios E.D., Griffin L.D., Lippincott-Schwartz J., Marsh M. (2014) **Flat clathrin lattices: stable features of the plasma membrane** *Mol Biol Cell* **25**:3581–3594 [Google Scholar](#)
- Hamdan F.F., Myers C.T., Cossette P., Lemay P., Spiegelman D., Laporte A.D., Nassif C., Diallo O., Monlong J., Cadieux-Dion M., Dobrzyniecka S., Meloche C., Retterer K., Cho M.T., Rosenfeld J.A., Bi W., Massicotte C., Miguët M., Brunga L., Regan B.M., Mo K., Tam C., Schneider A., Hollingsworth G., FitzPatrick D.R., Donaldson A., Canham N., Blair E., Kerr B., Fry A.E., Thomas R.H., Shelagh J., Hurst J.A., Brittain H., Blyth M., Lebel R.R., Gerkes E.H., Davis-Keppen L., Stein Q., Chung W.K., Dorison S.J., Benke P.J., Fassi E., Corsten-Janssen N., Kamsteeg E.-J., Mau-Them F.T., Bruel A.-L., Verloes A., Öunap K., Wojcik M.H., Albert D.V.F., Venkateswaran S., Ware T., Jones D., Liu Y.-C., Mohammad S.S., Bizargity P., Bacino C.A., Leuzzi V., Martinelli S., Dallapiccola B., Tartaglia M., Blumkin L., Wierenga K.J., Purcarin G., O’Byrne J.J., Stockler S., Lehman A., Keren B., Nougues M.-C., Mignot C., Auvin S., Nava C., Hiatt S.M., Bebin M., Shao Y., Scaglia F., Lalani S.R., Frye R.E., Jarjour I.T., Jacques S., Boucher R.-M., Riou E., Srouf M., Carmant L., Lortie A., Major P., Diadori P., Dubeau F., D’Anjou G., Bourque G., Berkovic S.F., Sadleir L.G., Campeau P.M., Kibar Z., Lafrenière R.G., Girard S.L., Mercimek-Mahmutoglu S., Boelman C., Rouleau G.A.,

et al. (2017) **High Rate of Recurrent De Novo Mutations in Developmental and Epileptic Encephalopathies** *The American Journal of Human Genetics* **101**:664–685 [Google Scholar](#)

Harel T., Griffin J.N., Arbogast T., Monroe T.O., Palombo F., Martinelli M., Seri M., Pippucci T., Elpeleg O., Katsanis N. (2020) **Loss of function mutations in CCDC32 cause a congenital syndrome characterized by craniofacial, cardiac and neurodevelopmental anomalies** *Human Molecular Genetics* **29**:1489–1497 [Google Scholar](#)

Helbig I., Lopez-Hernandez T., Shor O., Galer P., Ganesan S., Pendziwiat M., Rademacher A., Ellis C.A., Hümpfer N., Schwarz N., Seiffert S., Peeden J., Shen J., Štěrbová K., Hammer T.B., Møller R.S., Shinde D.N., Tang S., Smith L., Poduri A., Krause R., Benninger F., Helbig K.L., Haucke V., Weber Y.G. (2019) **A Recurrent Missense Variant in AP2M1 Impairs Clathrin-Mediated Endocytosis and Causes Developmental and Epileptic Encephalopathy** *Am J Hum Genet* **104**:1060–1072 [Google Scholar](#)

Henne W.M., Boucrot E., Meinecke M., Evergren E., Vallis Y., Mittal R., McMahon H.T. (2010) **FCHo proteins are nucleators of clathrin-mediated endocytosis** *Science* **328**:1281–1284 [Google Scholar](#)

Jackson L.P., Kelly B.T., McCoy A.J., Gaffry T., James L.C., Collins B.M., Höning S., Evans P.R., Owen D.J. (2010) **A large-scale conformational change couples membrane recruitment to cargo binding in the AP2 clathrin adaptor complex** *Cell* **141**:1220–1229 [Google Scholar](#)

Jaqaman K., Loerke D., Mettlen M., Kuwata H., Grinstein S., Schmid S.L., Danuser G. (2008) **Robust single-particle tracking in live-cell time-lapse sequences** *Nat Methods* **5**:695–702 [Google Scholar](#)

Jumper J., Evans R., Pritzel A., Green T., Figurnov M., Ronneberger O., Tunyasuvunakool K., Bates R., Žídek A., Potapenko A., Bridgland A., Meyer C., Kohl S.A.A., Ballard A.J., Cowie A., Romera-Paredes B., Nikolov S., Jain R., Adler J., Back T., Petersen S., Reiman D., Clancy E., Zielinski M., Steinegger M., Pacholska M., Berghammer T., Bodenstein S., Silver D., Vinyals O., Senior A.W., Kavukcuoglu K., Kohli P., Hassabis D. (2021) **Highly accurate protein structure prediction with AlphaFold** *Nature* **596**:583–589 [Google Scholar](#)

Jung S., Maritzen T., Wichmann C., Jing Z., Neef A., Revelo N.H., Al-Moyed H., Meese S., Wojcik S.M., Panou I., Bulut H., Schu P., Ficner R., Reisinger E., Rizzoli S.O., Neef J., Strenzke N., Haucke V., Moser T. (2015) **Disruption of adaptor protein 2μ (AP-2μ) in cochlear hair cells impairs vesicle reloading of synaptic release sites and hearing** *EMBO J* **34**:2686–2702 [Google Scholar](#)

Kaksonen M., Roux A. (2018) **Mechanisms of clathrin-mediated endocytosis** *Nature Reviews Molecular Cell Biology* **19**:313–326 [Google Scholar](#)

Kirchhausen T., Owen D., Harrison S.C. (2014) **Molecular structure, function, and dynamics of clathrin-mediated membrane traffic** *Cold Spring Harb Perspect Biol* **6**:a016725 [Google Scholar](#)

Kutner R.H., Zhang X.Y., Reiser J. (2009) **Production, concentration and titration of pseudotyped HIV-1-based lentiviral vectors** *Nature Protocols* **4**:495–505 [Google Scholar](#)

Li B., Ma L., Zhang C., Zhou Z., Yuan H., Jiang H., Pan Y., Tan Q. (2018) **Associations of genetic variants in endocytic trafficking of epidermal growth factor receptor super pathway with**

risk of nonsyndromic cleft lip with or without cleft palate *Mol Genet Genomic Med* **6**:1157–1167 [Google Scholar](#)

Li Q., Laumonnier Y., Syrovets T., Simmet T. (2011) **Yeast two-hybrid screening of proteins interacting with plasmin receptor subunit: C-terminal fragment of annexin A2** *Acta Pharmacol Sin* **32**:1411–1418 [Google Scholar](#)

Li W., Puertollano R., Bonifacio J.S., Overbeek P.A., Everett E.T. (2010) **Disruption of the murine Ap2 β 1 gene causes nonsyndromic cleft palate** *Cleft Palate Craniofac J* **47**:566–573 [Google Scholar](#)

Loerke D., Mettlen M., Schmid S.L., Danuser G. (2011) **Measuring the Hierarchy of Molecular Events During Clathrin-Mediated Endocytosis** *Traffic* **12**:815–825 [Google Scholar](#)

Manti F., Nardecchia F., Barresi S., Venditti M., Pizzi S., Hamdan F.F., Blau N., Burlina A., Tartaglia M., Leuzzi V. (2019) **Neurotransmitter trafficking defect in a patient with clathrin (CLTC) variation presenting with intellectual disability and early-onset parkinsonism** *Parkinsonism & Related Disorders* **61**:207–210 [Google Scholar](#)

McMahon H.T., Boucrot E. (2011) **Molecular mechanism and physiological functions of clathrin-mediated endocytosis** *Nat Rev Mol Cell Biol* **12**:517–533 [Google Scholar](#)

Meng E.C., Goddard T.D., Pettersen E.F., Couch G.S., Pearson Z.J., Morris J.H., Ferrin T.E. (2023) **UCSF ChimeraX: Tools for structure building and analysis** *Protein Science* **32**:e4792 [Google Scholar](#)

Merrifield C.J., Kaksonen M. (2014) **Endocytic Accessory Factors and Regulation of Clathrin-Mediated Endocytosis** *Cold Spring Harbor Perspectives in Biology* **6** [Google Scholar](#)

Messa M., Fernández-Busnadiego R., Sun E.W., Chen H., Czapla H., Wrasman K., Wu Y., Ko G., Ross T., Wendland B., De Camilli P. (2014) **Epsin deficiency impairs endocytosis by stalling the actin-dependent invagination of endocytic clathrin-coated pits** *eLife* **3**:e03311 <https://doi.org/10.7554/eLife.03311> | [Google Scholar](#)

Mettlen M., Chen P.H., Srinivasan S., Danuser G., Schmid S.L. (2018) **Regulation of Clathrin-Mediated Endocytosis** *Annu Rev Biochem* **87**:871–896 [Google Scholar](#)

Mettlen M., Danuser G. (2014) **Imaging and Modeling the Dynamics of Clathrin-Mediated Endocytosis** *Cold Spring Harbor Perspectives in Biology* **6** [Google Scholar](#)

Mino R.E., Chen Z., Mettlen M., Schmid S.L. (2020) **An internally eGFP-tagged α -adaptin is a fully functional and improved fiduciary marker for clathrin-coated pit dynamics** *Traffic* **21**:603–616 [Google Scholar](#)

Olesen L.E., Ford M.G.J., Schmid E.M., Vallis Y., Babu M.M., Li P.H., Mills I.G., McMahon H.T., Praefcke G.J.K. (2008) **Solitary and Repetitive Binding Motifs for the AP2 Complex α -Appendage in Amphiphysin and Other Accessory Proteins*** *Journal of Biological Chemistry* **283**:5099–5109 [Google Scholar](#)

Praefcke G.J., Ford M.G., Schmid E.M., Olesen L.E., Gallop J.L., Peak-Chew S.Y., Vallis Y., Babu M.M., Mills I.G., McMahon H.T. (2004) **Evolving nature of the AP2 α -appendage hub during clathrin-coated vesicle endocytosis** *EMBO J* **23**:4371–4383 [Google Scholar](#)

- Reis C.R., Chen P.H., Srinivasan S., Aguet F., Mettlen M., Schmid S.L. (2015) **Crosstalk between Akt/GSK3 β signaling and dynamin-1 regulates clathrin-mediated endocytosis** *EMBO J* **34**:2132–2146 [Google Scholar](#)
- Saffarian S., Kirchhausen T. (2008) **Differential evanescence nanometry: live-cell fluorescence measurements with 10-nm axial resolution on the plasma membrane** *Biophys J* **94**:2333–2342 [Google Scholar](#)
- Schmid E.M., Ford M.G.J., Burtey A., Praefcke G.J.K., Peak-Chew S.-Y., Mills I.G., Benmerah A., McMahon H.T. (2006) **Role of the AP2 β -Appendage Hub in Recruiting Partners for Clathrin-Coated Vesicle Assembly** *PLOS Biology* **4**:e262 [Google Scholar](#)
- Schmid S.L., Smythe E. (1991) **Stage-specific assays for coated pit formation and coated vesicle budding in vitro** *J Cell Biol* **114**:869–880 [Google Scholar](#)
- Smith S.M., Smith C.J. (2022) **Capturing the mechanics of clathrin-mediated endocytosis** *Curr Opin Struc Biol* **75**:102427 [Google Scholar](#)
- Sochacki K.A., Dickey A.M., Strub M.-P., Taraska J.W. (2017) **Endocytic proteins are partitioned at the edge of the clathrin lattice in mammalian cells** *Nature Cell Biology* **19**:352–361 [Google Scholar](#)
- Srinivasan S., Burckhardt C.J., Bhawe M., Chen Z., Chen P.H., Wang X., Danuser G., Schmid S.L. (2018) **A noncanonical role for dynamin-1 in regulating early stages of clathrin-mediated endocytosis in non-neuronal cells** *PLoS Biol* **16**:e2005377 [Google Scholar](#)
- Svitkina T (2022) **Imaging Cytoskeleton Components by Electron Microscopy** *Methods Mol Biol* **2364**:25–52 [Google Scholar](#)
- Svitkina T.M (2017) **Platinum replica electron microscopy: Imaging the cytoskeleton globally and locally** *Int J Biochem Cell Biol* **86**:37–41 [Google Scholar](#)
- Sznajder Ł.J., Swanson M.S. (2019) **Short Tandem Repeat Expansions and RNA-Mediated Pathogenesis in Myotonic Dystrophy** *International Journal of Molecular Sciences* **20**:3365 [Google Scholar](#)
- Tagiltsev G., Haselwandter C.A., Scheuring S. (2021) **Nanodissected elastically loaded clathrin lattices relax to increased curvature** *Science Advances* **7** [Google Scholar](#)
- Taylor M.J., Perrais D., Merrifield C.J. (2011) **A High Precision Survey of the Molecular Dynamics of Mammalian Clathrin-Mediated Endocytosis** *PLOS Biology* **9**:e1000604 [Google Scholar](#)
- Traub L.M (2011) **Regarding the Amazing Choreography of Clathrin Coats** *Plos Biology* **9** [Google Scholar](#)
- Umasankar P.K., Ma L., Thieman J.R., Jha A., Doray B., Watkins S.C., Traub L.M. (2014) **A clathrin coat assembly role for the muniscin protein central linker revealed by TALEN-mediated gene editing** *eLife* **3** [Google Scholar](#)

Umasankar P.K., Sanker S., Thieman J.R., Chakraborty S., Wendland B., Tsang M., Traub L.M. (2012) **Distinct and separable activities of the endocytic clathrin-coat components Fcho1/2 and AP-2 in developmental patterning** *Nat Cell Biol* **14**:488–501 [Google Scholar](#)

Varadi M., Anyango S., Deshpande M., Nair S., Natassia C., Yordanova G., Yuan D., Stroe O., Wood G., Laydon A., Židek A., Green T., Tunyasuvunakool K., Petersen S., Jumper J., Clancy E., Green R., Vora A., Lutfi M., Figurnov M., Cowie A., Hobbs N., Kohli P., Kleywegt G., Birney E., Hassabis D., Velankar S. (2021) **AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models** *Nucleic Acids Research* **50**:D439–D444 [Google Scholar](#)

Wainberg M., Kamber R.A., Balsubramani A., Meyers R.M., Sinnott-Armstrong N., Hornburg D., Jiang L., Chan J., Jian R., Gu M., Shcherbina A., Dubreuil M.M., Spees K., Meuleman W., Snyder M.P., Bassik M.C., Kundaje A. (2021) **A genome-wide atlas of co-essential modules assigns function to uncharacterized genes** *Nature Genetics* **53**:638–649 [Google Scholar](#)

Wan C., Puscher H., Ouyang Y., Wu J., Tian Y., Li S., Yin Q., Shen J. (2024) **An AAGAB-to-CCDC32 handover mechanism controls the assembly of the AP2 adaptor complex** *Proc Natl Acad Sci U S A* **121**:e2409341121 [Google Scholar](#)

Wang X., Chen Z., Mettlen M., Noh J., Schmid S.L., Danuser G. (2020) **DASC, a sensitive classifier for measuring discrete early stages in clathrin-mediated endocytosis** *eLife* **9**:e53686 <https://doi.org/10.7554/eLife.53686> | [Google Scholar](#)

Wu F., Yao P.J. (2009) **Clathrin-mediated endocytosis and Alzheimer's disease: An update** *Ageing Res Rev* **8**:147–149 [Google Scholar](#)

Yang C., Colosi P., Hugelier S., Zabezhinsky D., Lakadamyali M., Svitkina T. (2022) **Actin polymerization promotes invagination of flat clathrin-coated lattices in mammalian cells by pushing at lattice edges** *Nature Communications* **13**:6127 [Google Scholar](#)

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Reviewer #1 (Public review):

This is a revision of a manuscript previously submitted to Review Commons. The authors have partially addressed my comments, mainly by expanding the introduction and discussion sections. Sandy Schmid, a leading expert on the AP2 adaptor and CME, has been added as a co-corresponding author. The main message of the manuscript remains unchanged. Through overexpression of fluorescently tagged CCDC32, the authors propose that, in addition to its established role in AP2 assembly, CCDC32 also follows AP2 to the plasma membrane and regulates CCP maturation. The manuscript presents some interesting ideas, but there are still concerns regarding data inconsistencies and gaps in the evidence.

(1) eGFP-CCDC32 was expressed at 5-10 times higher levels than endogenous CCDC32. This high expression can artificially drive CCDC32 to the cell surface via binding to the alpha appendage domain (AD)-an interaction that may not occur under physiological conditions.

(2) Which region of CCDC32 mediates alpha AD binding? Strangely, the only mutant tested in this work, $\Delta 78-98$, still binds AP2, but shifts to binding only mu and beta. If the authors claim that CCDC32 is recruited to mature AP2 via the alpha AD, then a mutant deficient in alpha AD binding should not bind AP2 at all. Such a mutant is critical for establish the model proposed in this work.

(3) The concept of hemicomplexes is introduced abruptly. What is the evidence that such hemicomplexes exist? If CCDC32 binds to hemicomplexes, this must occur in the cytosol, as only mature AP2 tetramers are recruited to the plasma membrane. The authors state that CCDC32 binds the AD of alpha but not beta, so how can the $\Delta 78-98$ mutant bind mu and beta?

(4) The reported ability of CCDC32 to pull down AP2 beta is puzzling. Beta is not found in the CCDC32 interactome in two independent studies using 293 and HCT116 cells (BioPlex). In addition, clathrin is also absent in the interactome of CCDC32, which is difficult to reconcile with a proposed role in CCPs. Can the authors detect CCDC32 binding to clathrin?

(5) Figure 5B appears unusual-is this a chimera? Figure 5C likely reflects a mixture of immature and mature AP2 adaptor complexes.

(6) CCDC32 is reduced by about half in siRNA knockdown. Why not use CRISPR to completely eliminate CCDC32 expression?

<https://doi.org/10.7554/eLife.107039.1.sa3>

Reviewer #2 (Public review):

Yang et al. describes CCDC32 as a new clathrin mediated endocytosis (CME) accessory protein. The authors show that CCDC32 binds directly to AP2 via a small alpha helical region and cells depleted for this protein show defective CME. Finally, the authors show that the CCDC32

nonsense mutations found in patients with cardio-facial-neuro-developmental syndrome (CFNDS) disrupt the interaction of this protein to the AP2 complex. The results presented suggest that CCDC32 may act as both a chaperone (as recently published) and a structural component of the AP2 complex.

Strengths:

The conclusions presented are generally well supported by experimental data and the authors carefully point out the differences between their results and the results by Wan et al. (PNAS 2024).

Weaknesses:

The experiments regarding the role of CCDC32 in CFNDS still require some clarifications to make them clearer to scientists working on this disease. The authors fail to describe that the CCDC32 isoform they use in their studies is different from the one used when CFNDS patient mutations were described. This may create some confusion. Also, the authors did not discuss that the frame-shift mutations in patients may be leading to nonsense mediated decay.

<https://doi.org/10.7554/eLife.107039.1.sa2>

Reviewer #3 (Public review):

In this manuscript, Yang et al. characterize the endocytic accessory protein CCDC32, which has implications in cardio-facio-neuro-developmental syndrome (CFNDS). The authors clearly demonstrate that the protein CCDC32 has a role in the early stages of endocytosis, mainly through the interaction with the major endocytic adaptor protein AP2, and they identify regions taking part in this recognition. Through live cell fluorescence imaging and electron microscopy of endocytic pits, the authors characterize the lifetimes of endocytic sites, the formation rate of endocytic sites and pits and the invagination depth, in addition to transferrin receptor (TfnR) uptake experiments. Binding between CCDC32 and CCDC32 mutants to the AP2 alpha appendage domain is assessed by pull down experiments. While interaction between CCDC32 and the alpha appendage domain of AP2 is clearly described, a discussion of potential association with other AP2 domains would be beneficial to understand the impact of CCDC32 in endocytosis.

Together, these experiments allow deriving a phenotype of CCDC32 knock-down and CCDC32 mutants within endocytosis, which is a very robust system, in which defects are not so easily detected. A mutation of CCDC32, mimicking CFNDS mutations, is also addressed in this study and shown to have endocytic defects.

In summary, the authors present a strong combination of techniques, assessing the impact of CCDC32 in clathrin mediated endocytosis and its binding to AP2.

<https://doi.org/10.7554/eLife.107039.1.sa1>

Author response:

(1) General Statements

As you will see in our attached rebuttal to the reviewers, we have added several new experiments and revised manuscript to fully address their concerns.

(2) Point-by-point description of the revisions

Reviewer #1:

Evidence, reproducibility and clarity

Summary:

The manuscript by Yang et al. describes a new CME accessory protein. CCDC32 has been previously suggested to interact with AP2 and in the present work the authors confirm this interaction and show that it is a bona fide CME regulator. In agreement with its interaction with AP2, CCDC32 recruitment to CCPs mirrors the accumulation of clathrin. Knockdown of CCDC32 reduces the amount of productive CCPs, suggestive of a stabilisation role in early clathrin assemblies. Immunoprecipitation experiments mapped the interaction of CCDC42 to the α -appendage of the AP2 complex α -subunit. Finally, the authors show that the CCDC32 nonsense mutations found in patients with cardio-facial-neuro-developmental syndrome disrupt the interaction of this protein to the AP2 complex. The manuscript is well written and the conclusions regarding the role of CCDC32 in CME are supported by good quality data. As detailed below, a few improvements/clarifications are needed to reinforce some of the conclusions, especially the ones regarding CFNDS.

We thank the referee for their positive comments. In light of a recently published paper describing CCDC32 as a co-chaperone required for AP2 assembly (Wan et al., PNAS, 2024, see reviewer 2), we have added several additional experiments to address all concerns and consequently gained further insight into CCDC32-AP2 interactions and the important dual role of CCDC32 in regulating CME.

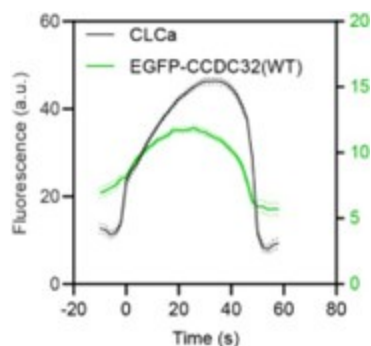
Major comments:

(1) Why did the protein could just be visualized at CCPs after knockdown of the endogenous protein? This is highly unusual, especially on stable cell lines. Could this be that the tag is interfering with the expressed protein function rendering it incapable of outcompeting the endogenous? Does this points to a regulated recruitment?

The reviewer is correct, this would be unusual; however, it is not the case. We misspoke in the text (although the figure legend was correct) these experiments were performed without siRNA knockdown and we can indeed detect eGFP-CCDC32 being recruited to CCPs in the presence of endogenous protein. Nonetheless, we repeated the experiment to be certain (see Author response image 1).

Author response image 1.

Cohort-averaged fluorescence intensity traces of CCPs (marked with mRuby-CLCa) and CCP-enriched eGFPCCDC32(FL).



*(2) The disease mutation used in the paper does not correspond to the truncation found in patients. The authors use an 1-54 truncation, but the patients described in Harel et al. have frame shifts at the positions 19 (Thr19Tyrfs*12) and 64 (Glu64Glyfs*12), while the patient described in Abdalla et al. have the deletion of two introns, leading to a frameshift around amino acid 90. Moreover, to be precisely test the function of these disease mutations, one would need to add the extra amino acids generated by the frame shift. For example, as denoted in the mutation description in Harel et al., the frameshift at position 19 changes the Threonine 19 to a Tyrosine and adds a run of 12 extra amino acids (Thr19Tyrfs*12).*

The label of the disease mutant p.(Thr19Tyrfs12) and p.(Glu64Glyfs12) is based on a 194aa polypeptide version of CCDC32 initiated at a nonconventional start site that contains a 9 aa peptide (VRGSLRFQ) upstream of the N-terminus we show. Thus, we are indeed using the appropriate mutation site (see: <https://www.uniprot.org/uniprotkb/Q9BV29/entry>). The reviewer is correct that we have not included the extra 12 aa in our construct; however as these residues are not present in the other CFNDS mutants, we think it unlikely that they contribute to the disease phenotype. Rather, as neither of the clinically observed mutations contain the 78-98 aa sequence required for AP2 binding and CME function, we are confident that this defect contributed to the disease. Thus, we are including the data on the CCDC32(1-54) mutant, as we believe these results provide a valuable physiological context to our studies.

(3) The frameshift caused by the CFNDS mutations (especially the one studied) will likely lead to nonsense mediated RNA decay (NMD). The frameshift is well within the rules where NMD generally kicks in. Therefore, I am unsure about the functional insights of expressing a diseaserelated protein which is likely not present in patients.

We thank the reviewer for bringing up this concern. However, as shown in new Figure S1, the mutant protein is expressed at comparable levels as the WT, suggesting that NMD is not occurring.

(4) Coiled coils generally form stable dimers. The typically hydrophobic core of these structures is not suitable for transient interactions. This complicates the interpretation of the results regarding the role of this region as the place where the interaction to AP2 occurs. If the coiled coil holds a stable CCDC32 dimer, disrupting this dimer could reduce the affinity to AP2 (by reduced avidity) to the actual binding site. A construct with an orthogonal dimeriser or a pulldown of the delta78-98 protein with of the GST AP2a-AD could be a good way to sort this issue.

We were unable to model a stable dimer (or other oligomer) of this protein with high confidence using AlphaFold 3.0. Moreover, we were unable to detect endogenous CCDC32 coimmunoprecipitating with eGFP-CCDC32 (Fig. S6C). Thus, we believe that the moniker, based solely on the alpha-helical content of the protein is a misnomer. We have explained this in the main text.

Minor comments:

(1) The authors interchangeably use the term "flat CCPs" and "flat clathrin lattices". While these are indeed related, flat clathrin lattices have been also used to refer to "clathrin plaques". To avoid confusion, I suggest sticking to the term "flat CCPs" to refer to the CCPs which are in their early stages of maturation.

Agreed. Thank you for the suggestion. We have renamed these structures flat clathrin assemblies, as they do not acquire the curvature needed to classify them as pits, and do not grow to the size that would classify them as plaques.

Significance

General assessment:

CME drives the internalisation of hundreds of receptors and surface proteins in practically all tissues, making it an essential process for various physiological processes. This versatility comes at the cost of a large number of molecular players and regulators. To understand this complexity, unravelling all the components of this process is vital. The manuscript by Yang et al. gives an important contribution to this effort as it describes a new CME regulator, CCDC32, which acts directly at the main CME adaptor AP2. The link to disease is interesting, but the authors need to refine their experiments. The requirement for endogenous knockdown for recruitment of the tagged CCDC32 is unusual and requires further exploration.

Advance:

The increased frequency of abortive events presented by CCDC32 knockdown cells is very interesting, as it hints to an active mechanism that regulates the stabilisation and growth of clathrin coated pits. The exact way clathrin coated pits are stabilised is still an open question in the field.

Audience:

This is a basic research manuscript. However, given the essential role of CME in physiology and the growing number of CME players involved in disease, this manuscript can reach broader audiences.

We thank the referee for recognizing the ‘interesting’ advances our studies have made and for considering these studies as ‘an important contribution’ to ‘an essential process for various physiological processes’ and able ‘to reach broader audiences’. We have addressed and reconciled the reviewer’s concerns in our revised manuscript.

Field of expertise of the reviewer:

Clathrin mediated endocytosis, cell biology, microscopy, biochemistry.

Reviewer #2:

Evidence, reproducibility and clarity

In this manuscript, the authors demonstrate that CCDC32 regulates clathrin-mediated endocytosis (CME). Some of the findings are consistent with a recent report by Wan et al. (2024 PNAS), such as the observation that CCDC32 depletion reduces transferrin uptake and diminishes the formation of clathrin-coated pits. The primary function of CCDC32 is to regulate AP2 assembly, and its depletion leads to AP2 degradation. However, this study did not examine AP2 expression levels. CCDC32 may bind to the appendage domain of AP2 alpha, but it also binds to the core domain of AP2 alpha.

We thank the reviewer for drawing our attention to the Wan et al. paper, that appeared while this work was under review. However, our in vivo data are not fully consistent with the report from Wan et al. The discrepancies reveal a dual function of CCDC32 in CME that was

masked by complete knockout vs siRNA knockdown of the protein, and also likely affected by the position of the GFP-tag (C- vs N-terminal) on this small protein. Thus:

- Contrary to Wan et al., we do not detect any loss of AP2 expression (see new Figure S3A-B) upon siRNA knockdown. Most likely the ~40% residual CCDC32 present after siRNA knockdown is sufficient to fulfill its catalytic chaperone function but not its structural role in regulating CME beyond the AP2 assembly step.
- Contrary to Wan et al., we have shown that CCDC32 indeed interacts with intact AP2 complex (Figure S3C and 6B,C) showing that all 4 subunits of the AP2 complex co-IP with full length eGFP-CCDC32. Interestingly, whereas the full length CCDC32 pulls down the intact AP2 complex, co-IP of the $\Delta 78-98$ mutant retains its ability to pull down the $\beta 2$ - $\mu 2$ hemicomplex, its interactions with $\alpha : \sigma 2$ are severely reduced. While this result is consistent with the report of Wan et al that CCDC32 binds to the $\alpha : \sigma 2$ hemi-complex, it also suggests that the interactions between CCDC32 and AP2 are more complex and will require further studies.
- Contrary to Wan et al., we provide strong evidence that CCDC32 is recruited to CCPs. Interestingly, modeling with AlphaFold 3.0 identifies a highly probable interaction between alpha helices encoded by residues 66-91 on CCDC32 and residues 418-438 on α . The latter are masked by $\mu 2$ -C in the closed confirmation of the AP2 core, but exposed in the open confirmation triggered by cargo binding, suggesting that CCDC32 might only bind to membrane-bound AP2.

Thus, our findings are indeed novel and indicate striking multifunctional roles for CCDC32 in CME, making the protein well worth further study.

(1) Besides its role in AP2 assembly, CCDC32 may potentially have another function on the membrane. However, there is no direct evidence showing that CCDC32 associates with the plasma membrane.

We disagree, our data clearly shows that CCDC32 is recruited to CCPs (Fig. 1B) and that CCPs that fail to recruit CCDC32 are short-lived and likely abortive (Fig. 1C). Wan et al. did not observe any colocalization of C-terminally tagged CCDC32 to CCPs, whereas we detect recruitment of our N-terminally tagged construct, which we also show is functional (Fig. 6F). Further, we have demonstrated the importance of the C-terminal region of CCDC32 in membrane association (see new Fig. S7). Thus, we speculate that a C-terminally tagged CCDC32 might not be fully functional. Indeed, SIM images of the C-terminally-tagged CCDC32 in Wan et al., show large (~100 nm) structures in the cytosol, which may reflect aggregation.

(2) CCDC32 binds to multiple regions on AP2, including the core domain. It is important to distinguish the functional roles of these different binding sites.

We have localized the AP2-ear binding region to residues 78-99 and shown these to be critical for the functions we have identified. As described above we now include data that are complementary to those of Wan et al. However, our data also clearly points to additional binding modalities. We agree that it will be important to map these additional interactions and identify their functional roles, but this is beyond the scope of this paper.

(3) AP2 expression levels should be examined in CCDC32 depleted cells. If AP2 is gone, it is not surprising that clathrin-coated pits are defective.

Agreed and we have confirmed this by western blotting (Figure S3A-B) and detect no reduction in levels of any of the AP2 subunits in CCDC32 siRNA knockdown cells. As stated above this could be due to residual CCDC32 present in the siRNA KD vs the CRISPR-mediated gene KO.

(4) If the authors aim to establish a secondary function for CCDC32, they need to thoroughly discuss the known chaperone function of CCDC32 and consider whether and how CCDC32 regulates a downstream step in CME.

Agreed. We have described the Wan et al paper, which came out while our manuscript was in review, in our Introduction. As described above, there are areas of agreement and of discrepancies, which are thoroughly documented and discussed throughout the revised manuscript.

(5) The quality of Figure 1A is very low, making it difficult to assess the localization and quantify the data.

The low signal:noise in Fig. 1A the reviewer is concerned about is due to a diffuse distribution of CCDC32 on the inner surface of the plasma membrane. We now, more explicitly describe this binding, which we believe reflects a specific interaction mediated by the C-terminus of CCDC32; thus the degree of diffuse membrane binding we observe follows: eGFP-CCDC32(FL)> eGFPCCDC32(Δ 78-98)>eGFP-CCDC32(1-54)~eGFP/background (see new Fig. S7). Importantly, the colocalization of CCDC32 at CCPs is confirmed by the dynamic imaging of CCPs (Fig 1B).

(6) In Figure 6, why aren't AP2 mu and sigma subunits shown?

Agreed. Not being aware of CCDC32's possible dual role as a chaperone, we had assumed that the AP2 complex was intact. We have now added this data in Figure 6 B,C and Fig. S3C, as discussed above.

Page 5, top, this sentence is confusing: "their surface area ($\sim 17 \times 10 \text{ nm}^2$) remains significantly less than that required for the average 100 nm diameter CCV ($\sim 3.2 \times 10^3 \text{ nm}^2$)."

Thank you for the criticism. We have clarified the sentence and corrected a typo, which would definitely be confusing. The section now reads, "While the flat CCSs we detected in CCDC32 knockdown cells were significantly larger than in control cells (Fig. 4D, mean diameter of 147 nm vs. 127 nm, respectively), they are much smaller than typical long-lived flat clathrin lattices ($d \geq 300 \text{ nm}$)(Grove et al., 2014). Indeed, the surface area of the flat CCSs that accumulate in CCDC32 KD cells (mean $\sim 1.69 \times 10^4 \text{ nm}^2$) remains significantly less than the surface area of an average 100 nm diameter CCV ($\sim 3.14 \times 10^4 \text{ nm}^2$). Thus, we refer to these structures as 'flat clathrin assemblies' because they are neither curved 'pits' nor large 'lattices'. Rather, the flat clathrin assemblies represent early, likely defective, intermediates in CCP formation."

Significance

Overall, while this work presents some interesting ideas, it remains unclear whether CCDC32 regulates AP2 beyond the assembly step.

Our responses above argue that we have indeed established that CCDC32 regulates AP2 beyond the assembly step. We have also identified several discrepancies between our findings and those reported by Wan et al., most notably binding between CCDC32 and mature AP2 complexes and the AP2-dependent recruitment of CCDC32 to CCPs. It is possible that these discrepancies may be due to the position of the GFP tag (ours is N-terminal, theirs is C-terminal; we show that the N-terminal tagged CCDC32 rescues the knockdown phenotype, while Wan et al., do not provide evidence for functionality of the C-terminal construct).

Reviewer #3:

Evidence, reproducibility and clarity (Required):

In this manuscript, Yang et al. characterize the endocytic accessory protein CCDC32, which has implications in cardio-facio-neuro-developmental syndrome (CFNDS). The authors clearly demonstrate that the protein CCDC32 has a role in the early stages of endocytosis, mainly through the interaction with the major endocytic adaptor protein AP2, and they identify regions taking part in this recognition. Through live cell fluorescence imaging and electron microscopy of endocytic pits, the authors characterize the lifetimes of endocytic sites, the formation rate of endocytic sites and pits and the invagination depth, in addition to transferrin receptor (TfnR) uptake experiments. Binding between CCDC32 and CCDC32 mutants to the AP2 alpha appendage domain is assessed by pull down experiments. Together, these experiments allow deriving a phenotype of CCDC32 knock-down and CCDC32 mutants within endocytosis, which is a very robust system, in which defects are not so easily detected. A mutation of CCDC32, known to play a role in CFNDS, is also addressed in this study and shown to have endocytic defects.

We thank the reviewer for their positive remarks regarding the quality of our data and the strength of our conclusions.

In summary, the authors present a strong combination of techniques, assessing the impact of CCDC32 in clathrin mediated endocytosis and its binding to AP2, whereby the following major and minor points remain to be addressed:

- The authors show that CCDC32 depletion leads to the formation of brighter and static clathrin coated structures (Figure 2), but that these were only prevalent to 7.8% and masked the 'normal' dynamic CCPs. At the same time, the authors show that the absence of CCDC32 induces pits with shorter life times (Figure 1 and Figure 2), the 'majority' of the pits.

Clarification is needed as to how the authors arrive at these conclusions and these numbers. The authors should also provide (and visualize) the corresponding statistics. The same statement is made again later on in the manuscript, where the authors explain their electron microscopy data. Was the number derived from there?

These points are critical to understanding CCDC32's role in endocytosis and is key to understanding the model presented in Figure 8. The numbers of how many pits accumulate in flat lattices versus normal endocytosis progression and the actual time scales could be included in this model and would make the figure much stronger.

Thank you for these comments. We understand the paradox between the visual impression and the reality of our dynamic measurements. We have been visually misled by this in previous work (Chen et al., 2020), which emphasizes the importance of unbiased image analysis afforded to us through the well-documented cmeAnalysis pipeline, developed by us (Aguet et al., 2013) and now used by many others (e.g. (He et al., 2020)).

The % of static structures was not derived from electron microscopy data, but quantified using cmeAnalysis, which automatically provides the lifetime distribution of CCPs. We have now clarified this in the manuscript and added a histogram (Fig. S4) quantifying the fraction of CCPs in lifetime cohorts <20s, 21-60s, 61-100s, 101-150s and >150s (static).

- In relation to the above point, the statistics of Figure 2E-G and the analysis leading there should also be explained in more detail: For example, what are the individual

points in the plot (also in Figures 6G and 7G)? The authors should also use a few phrases to explain software they use, for example DASC, in the main text.

Each point in these bar graphs represents a movie, where $n \geq 12$. These details have been added to the respective figure legend. We have also added a brief description of DASC analysis in the text.

- There are several questions related to the knock-down experiments that need to be addressed:

Firstly, knock-down of CCDC32 does not seem to be very strong (Figure S2B). Can the level of knock-down be quantified?

We have now quantified the KD efficiency. It is ~60%. This turns out to be fortuitous (see responses to reviewer 2), as a recent publication, which came out after we completed our study, has shown by CRISPR-mediated knockout, that CCDC32 also plays an essential chaperone function required for AP2 assembly. We do not see any reduction in AP2 levels or its complex formation under our conditions (see new Supplemental Figure S3), which suggests that the effects of CCDC32 on CCP dynamics are more sensitive to CCDC32 concentration than its roles as a chaperone. Our phenotypes would have been masked by more efficient depletion of CCDC32.

In page 6 it is indicated that the eGFP-CCDC32(1-54) and eGFP-CCDC32(Δ 78-98) constructs are siRNA-resistant. However in Fig S2B, these proteins do not show any signal in the western blot, so it is not clear if they are expressed or simply not detected by the antibody. The presence of these proteins after silencing endogenous CCDC32 needs to be confirmed to support Figures 6 and Figures 7, which critically rely on the presence of the CCDC32 mutants.

Unfortunately, the C-terminally truncated CCDC32 proteins are not detected because they lack the antibody epitope, indeed even the Δ 78-98 deletion is poorly detected (compare the GFP blot in new S1A with the anti-CCDC32 blot in S1B). However, these constructs contain the same siRNA-resistance mutation as the full length protein. That they are expressed and siRNA resistant can be seen in Fig. S2A (now Fig. S1A) blotting for GFP.

In Figures 6 and 7, siRNA knock-down of CCDC32 is only indicated for sub-figures F to G. Is this really the case? If not, the authors should clarify. The siRNA knock-down in Figure 1 is also only mentioned in the text, not in the figure legend. The authors should pay attention to make their figure legends easy to understand and unambiguous.

No, it is not the case. Thank you for pointing out the uncertainty. We have added these details to the Figure legends and checked all Figure legends to ensure that they clearly describe the data shown.

- It is not exactly clear how the curves in Figure 3C (lower panel) on the invagination depth were obtained. Can the authors clarify this a bit more? For example, what are k_T and k_E in Figure 3A? What is I_0 ? And how did the authors derive the logarithmic function used to quantify the invagination depth? In the main text, the authors say that the traces were 'logarithmically transformed'. This is not a technical term. The authors should refer to the actual equation used in the figure.

This analysis was developed by the Kirchhausen lab (Saffarian and Kirchhausen, 2008). We have added these details and reference them in the Figure legend and in the text. We also now use the more accurate descriptor 'log-transformed'.

- In the discussion, the claim 'The resulting dysregulation of AP2 inhibits CME, which further results in the development of CFNDS.' is maybe a bit too strong of a statement. Firstly, because the authors show themselves that CME is perturbed, but by no means inhibited. Secondly, the molecular link to CFNDS remains unclear. Even though CCDC32 mutants seem to be responsible for CFNDS and one of the mutant has been shown in this study to have a defect in endocytosis and AP2 binding, a direct link between CCDC32's function in endocytosis and CFNDS remains elusive. The authors should thus provide a more balanced discussion on this topic.

We have modified and softened our conclusions, which now read that the phenotypes we see likely “contribute to” rather than “cause” the disease.

- In Figure S1, the authors annotate the presence of a coiled-coil domain, which they also use later on in the manuscript to generate mutations. Could the authors specify (and cite) where and how this coiled-coil domain has been identified? Is this predicted helix indeed a coiled-coil domain, or just a helix, as indicated by the authors in the discussion?

See response to Reviewer 1, point 4. We have changed this wording to alpha-helix. The ‘coiled-coil’ reference is historical and unlikely a true reflection of CCDC32 structure. AlphaFold 3.0 predictions were unable to identify with certainty any coiled-coil structures, even if we modelled potential dimers or trimers; and we find no evidence of dimerization of CCDC32 *in vivo*. We have clarified this in the text.

Minor comments

- In general, a more detailed explanation of the microscopy techniques used and the information they report would be beneficial to provide access to the article also to non-expert readers in the field. This concerns particularly the analysis methods used, for example:

How were the cohort-averaged fluorescence intensity and lifetime traces obtained?

*How do the tools *cmeAnalysis* and *DASC* work? A brief explanation would be helpful.*

We have expanded Methods to add these details, and also described them in the main text.

- The axis label of Figure 2B is not quite clear. What does 'TfnR uptake % of surface bound' mean? Maybe the authors could explain this in more detail in the figure legend? Is the drop in uptake efficiency also accessible by visual inspection of the images? It would be interesting to see that.

This is a standard measure of CME efficiency. 'TfnR uptake % of surface bound' = Internalized TfnR/Surface bound TfnR. Again, images may be misleading as defects in CME lead to increased levels of TfnR on the cell surface, which in turn would result in more Tfn uptake even if the rate of CME is decreased.

- Figure 4: How is the occupancy of CCPs in the plasma membrane measured? What are the criteria used to divide CCSs into Flat, Dome or Sphere categories?

We have expanded Methods to add these details. Based on the degree of invagination, the shapes of CCSs were classified as either: flat CCSs with no obvious invagination; dome-shaped CCSs that had a hemispherical or less invaginated shape with visible edges of the clathrin lattice; and spherical CCSs that had a round shape with the invisible edges of clathrin lattice in 2D projection images. In most cases, the shapes were obvious in 2D PREM images. In

uncertain cases, the degree of CCS invagination was determined using images tilted at ± 10 – 20 degrees. The area of CCSs were measured using ImageJ and used for the calculation of the CCS occupancy on the plasma membrane.

- *Figure 5B: Can the authors explain, where exactly the GFP was engineered into AP2 alpha? This construct does not seem to be explained in the methods section.*

We have added this information. The construct, which corresponds to an insertion of GFP into the flexible hinge region of AP2, at aa649, was first described by (Mino et al., 2020) and shown to be fully functional. This information has been added to the Methods section.

- *Figure S1B: The authors should indicate the colour code used for the structural model.*

We have expanded our structural modeling using AlphaFold 3.0 in light of the recent publication suggesting the CCDC32 interacts with the $\mu 2$ subunit and does not bind full length AP2. These results are described in the text. The color coding now reflects certainty values given by AlphaFold 3.0 (Fig. S6B, D).

- *The list of primers referred to in the materials and methods section does not exist. There is a Table S1, but this contains different data. The actual Table S1 is not referenced in the main text. This should be done.*

We apologize for this error. We have now added this information in Table S2.

Significance (Required):

In this study, the authors analyse a so-far poorly understood endocytic accessory protein, CCDC32, and its implication for endocytosis. The experimental tool set used, allowing to quantify CCP dynamics and invagination is clearly a strength of the article that allows assessing the impact of an accessory protein towards the endocytic uptake mechanism, which is normally very robust towards mutations. Only through this detailed analysis of endocytosis progression could the authors detect clear differences in the presence and absence of CCDC32 and its mutants. If the above points are successfully addressed, the study will provide very interesting and highly relevant work allowing a better understanding of the early phases in CME with implication for disease.

The study is thus of potential interest to an audience interested in CME, in disease and its molecular reasons, as well as for readers interested in intrinsically disordered proteins to a certain extent, claiming thus a relatively broad audience. The presented results may initiate further studies of the so-far poorly understood and less well known accessory protein CCDC32.

We thank the reviewer for their positive comments on the significance of our findings and the importance of our detailed phenotypic analysis made possible by quantitative live cell microscopy. We also believe that our new structural modeling of CCDC32 and our findings of complex and extensive interactions with AP2 make the reviewers point regarding intrinsically disordered proteins even more interesting and relevant to a broad audience. We trust that our revisions indeed address the reviewer's concerns.

The field of expertise of the reviewer is structural biology, biochemistry and clathrin mediated endocytosis. Expertise in cell biology is rather superficial.

References:

- Aguet, F., Costin N. Antonescu, M. Mettlen, Sandra L. Schmid, and G. Danuser. 2013. Advances in Analysis of Low Signal-to-Noise Images Link Dynamin and AP2 to the Functions of an Endocytic Checkpoint. *Developmental Cell*. 26:279-291.
- Chen, Z., R.E. Mino, M. Mettlen, P. Michaely, M. Bhave, D.K. Reed, and S.L. Schmid. 2020. Wbox2: A clathrin terminal domain-derived peptide inhibitor of clathrin-mediated endocytosis. *Journal of Cell Biology*. 219.
- Grove, J., D.J. Metcalf, A.E. Knight, S.T. Wavre-Shapton, T. Sun, E.D. Protonotarios, L.D. Griffin, J. Lippincott-Schwartz, and M. Marsh. 2014. Flat clathrin lattices: stable features of the plasma membrane. *Mol Biol Cell*. 25:3581-3594.
- He, K., E. Song, S. Upadhyayula, S. Dang, R. Gaudin, W. Skillern, K. Bu, B.R. Capraro, I. Rapoport, I. Kusters, M. Ma, and T. Kirchhausen. 2020. Dynamics of Auxilin 1 and GAK in clathrin-mediated traffic. *J Cell Biol*. 219.
- Mino, R.E., Z. Chen, M. Mettlen, and S.L. Schmid. 2020. An internally eGFP-tagged α -adaptin is a fully functional and improved fiduciary marker for clathrin-coated pit dynamics. *Traffic*. 21:603-616.
- Saffarian, S., and T. Kirchhausen. 2008. Differential evanescence nanometry: live-cell fluorescence measurements with 10-nm axial resolution on the plasma membrane. *Biophys J*. 94:2333-2342.

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